Stochastic Origin Ensemble Reconstruction for Fast-Neutron Range Verification in Proton Therapy

by

Mohammad Hareth Hasan Al-Jomaa

Submitted to the Department of Computer Science, Electrical Engineering, and Mathematical Sciences
in partial fulfillment of the requirements for the degree of

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I hereby affirm that this work has been prepared independently, and that full and proper references to all sources employed have been provided, in accordance with the Regulations Relating to Studies and Examinations at the Western Norway University of Applied Sciences §12-1.

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"What we observe is not nature itself, but nature exposed to our method of questioning." [a]

— Werner Heisenberg

[a] *Physics and Philosophy*, 1958, p. 25.

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ABSTRACT

Proton therapy requires accurate knowledge of where the beam stops, but the delivered range is not directly observable during irradiation. Proton interactions, however, produce secondary radiation that can be measured directly and used to infer the range. This makes range verification an inverse problem: a problem in which unknown structures or physical quantities are inferred from their observed effects and measurements. In this setting, the detected secondary radiation provides the measurable evidence from which the range must be inferred. The Stochastic Origin Ensemble (SOE) is a reconstruction method that uses such measurements to infer possible origins of the detected secondary radiation.

This thesis formalises, implements, and evaluates whether the SOE applied to fast neutrons can produce a reliable range surrogate. The reconstructions are tested on PHITS-simulated list-mode data from monoenergetic proton pencil beams in a homogeneous water phantom. Reconstructed axial profiles are compared across beam energies of 100, 105, 110, 115, 120, 125, 135, and 150 MeV using peak-depth and R_{50} features calibrated against reference proton ranges. The reconstructed profiles do not preserve a reliable energy–depth relationship.

Before the SOE is applied, the detector geometry is found to compress the signal span over the eight energies by a factor of 1.73, leaving the reconstruction problem already constrained. The SOE then compresses the signal by a further factor of 1.88. As a result, the reconstructed features lose monotonicity, and the calibrated range errors reach the centimetre scale. Under the tested detector geometry, event statistics, voxelisation, and simulation assumptions, SOE reconstruction is not supported as a reliable basis for fast-neutron range-surrogate estimation. The work contributes a reproducible mathematical and computational implementation of SOE for fast-neutron cone data and identifies where future detector and reconstruction studies must preserve or recover axial range information.

Acknowledgments

First and foremost, praise be to God.

Isaac Newton once wrote that "If I have seen further it is by standing on the sholders of Giants."[1] I would not presume to compare myself to Newton, but the sentiment resonates. This thesis would not have been possible if it weren't for the giants who lifted me up to see; in my case, my supervisors, family, and friends.

I am grateful to Prof. Ilker Meric for trusting me with this problem and giving me the room to work through it at my own pace. His time, his feedback, and his continued involvement throughout are things I do not take for granted.

I am especially grateful to my supervisor, Dr. Hunter Ratliff, who remained closely engaged with the work even after moving to a new position. His comments consistently pressed me to think more carefully about what I was actually claiming, and where the limits of those claims really lay.

I am also deeply grateful to Asuman Kolbasi, who stepped in as supervisor during the final stages after Dr. Ratliff's departure. She gave generously of her time, offered clear direction when the work needed focus, and helped me bring the thesis to completion during the final period.

To my family and friends, thank you. Several of them sat patiently through explanations of work far outside their own interests, and their questions, often more searching than they realised, helped me find clearer ways to say what I had been struggling to articulate.

Finally, I regard the chance to have worked on this thesis as a great privilege, and I am grateful for it.

[1] Written in Newton's letter to Robert Hooke, dated 5 February 1675/1676.

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Chapter 1

Introduction

Cancer caused 9.7 million deaths worldwide in 2022 [1], and radiotherapy is part of the treatment pathway for approximately 52% of cancer patients [2]. Conventional photon radiotherapy deposits dose along the beam path and continues depositing dose beyond the target depth [3]. Charged-particle beams, including protons and carbon ions, have a well-defined range in tissue and produce a dose maximum near the end of their trajectory, although heavy-ion beams can also generate secondary fragments that contribute dose beyond the distal-fallof [4, 5]. For protons, this concentration occurs at the Bragg peak, the narrow depth interval where most of the proton energy is deposited before the beam stops [6].

This physical distinction is visible in the depth–dose comparison in Figure 1.1: photon dose decreases gradually with depth and continues beyond the high-dose region, whereas protons and carbon ions exhibit a finite range followed by a sharp distal fall-off. The same sharp fall-off that makes charged-particle therapy spatially selective also makes it sensitive to range error. The clinical value of proton therapy therefore depends on knowing where the Bragg peak and distal dose fall-off occur. A range error of even a few millimetres can shift the steep distal dose gradient into adjacent healthy tissue or out of the tumour, potentially overdosing organs at risk or underdosing the target [7]. By the end of 2023, more than 120 particle-therapy facilities were in clinical operation and approximately 350 000 patients had been treated with protons [8, 9]. This makes range accuracy a clinically relevant issue.

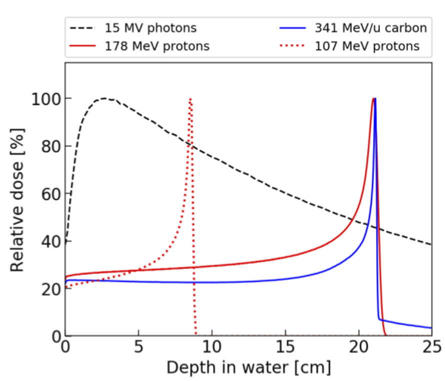


Figure 1.1: Illustrative depth–dose curves in water for photons, protons, and carbon ions. Adopted from [10]

The conversion from computed-tomography Hounsfield units to proton stopping-power ratio contributes materially to proton range uncertainty, with the magnitude depending on the imaging and calibration workflow [7]. The total uncertainty budget also includes patient setup variation, interfraction anatomical change, organ motion, and tissue heterogeneity along the beam path [11]. No single contribution is universal. Site-specific analyses yield total margins ranging from 2.8% of range plus 1.2 mm for liver and prostate to 6.3% plus 1.2 mm for lung, breast, and head-and-neck fields [12]. Clinical practice commonly applies a generic safety margin of approximately 3.5% of range plus 1 mm [7]. These margins ensure target coverage at the cost of excess dose to surrounding tissue: they compensate for range uncertainty but do not resolve it.

In standard clinical workflows, the delivered proton range is not directly observed during irradiation [13, 14]. Therapeutic proton beams are designed to deposit their energy inside the patient and to stop near the planned target depth [14]. The relevant range information is therefore internal, while the primary protons themselves are not directly imaged during delivery. Range verification consequently relies on secondary signals that can escape the patient and whose spatial distributions remain correlated with the proton path [13]. Nuclear interactions along the proton track generate secondary particles whose production profiles carry information about the proton path, including prompt gamma rays and fast neutrons [15, 16]. As illustrated in Figure 1.2, these secondary-particle profiles shift with beam energy, which makes their depth distributions relevant as range-surrogate signals.

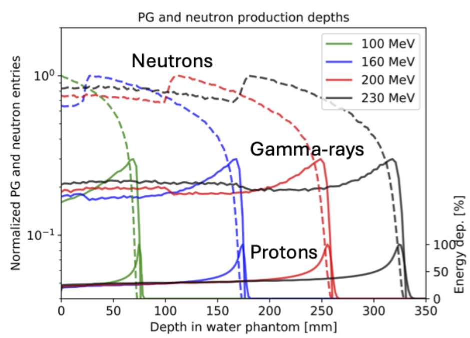


Figure 1.2: Illustrative depth profiles for prompt-gamma and fast-neutron production from proton beams in water, together with proton energy deposition. Adopted from [10]

This secondary-radiation approach provides the immediate context for the NOVO project, where prompt-gamma and fast-neutron measurements are investigated as indirect range-surrogate signals for proton-therapy verification.

1.1 NOVO

NOVO (*Next Generation Imaging for Real-Time Dose Verification Enabling Adaptive Proton Therapy*) is an European Innovation Council Pathfinder Open project running from 2024 to 2028 [10]. The project is coordinated by Western Norway University of Applied Sciences and brings together clinical, detector, electronics, modelling, and reconstruction expertise across an international consortium [10]. Its central aim is to develop proof-of-concept technology for real-time dose verification in proton therapy, with the longer-term goal of supporting dose-guided adaptive treatment.

NOVO addresses range verification through a hybrid multi-particle approach. Rather than relying on one secondary-radiation channel alone, the project aims to combine prompt-gamma and fast-neutron information with detector development, forward modelling, fast image reconstruction, and treatment-support methods [10]. The proposed detector concept uses organic scintillator elements with dual-ended photosensor readout, allowing detected interactions to provide time, position, energy, and particle-type information for reconstructing secondary-particle production distributions [17]. This dual use of prompt gammas and fast neutrons is central to the NOVO concept, since the two particle species carry complementary range-related information along the beam path [17].

Within the project, the thesis contributes to the reconstruction component of NOVO by focusing on fast-neutron cone data.

1.2 Problem statement

This study has three objectives:

1. To formalise and implement a modular SOE reconstruction pipeline for fast-neutron cone data.
2. To evaluate whether the reconstructed images preserve range-relevant structure when compared with reference information.
3. To document the computational performance and sampling behaviour of the implemented pipeline.

The evaluation asks whether the reconstructed distributions preserve a consistent energy–depth relationship, whether the derived range-surrogate metrics are stable enough to interpret, and whether the implemented pipeline executes the tested reconstructions within a documented computational cost. Together, these criteria assess whether the implemented SOE pipeline supports range-verification research under the tested conditions.

To the author's knowledge, prior SOE applications have not addressed fast-neutron cone data in the range-verification setting considered here [18–25].

1.3 Contributions

The main contribution is a reproducible mathematical and computational realisation of SOE for fast-neutron cone data. The thesis formalises the event-specific feasible surface, the event-indexed state space, the occupancy-based target, and the single-event Metropolis–Hastings kernel used by the implementation. These definitions make the implemented reconstruction pipeline explicit rather than leaving key choices implicit.

A second contribution is an explicit proposal construction for clipped cone surfaces. To the author's knowledge, the cited SOE literature does not state an Inverse Cumulative distribution function (inverse-CDF) construction for surface-area-uniform proposal sampling on such surfaces. This thesis records the construction used by the implementation and the proposal law required by the Metropolis–Hastings kernel.

A third contribution is the analysis workflow surrounding the reconstruction. The pipeline records outputs, runtime checks, sampling diagnostics, and timing measurements, while the geometry-only accessibility analysis tests how much axial information is present before SOE dynamics are applied. This separates limitations of the cone data from limitations introduced by the stochastic reconstruction itself.

The contribution is therefore not a new SOE principle, but a bounded formalisation, implementation, and diagnostic evaluation of SOE for fast-neutron range-verification data.

1.4 Scope

The study is conducted entirely in simulation. Proton transport, secondary-particle production, and detector interactions are simulated with the Particle and Heavy Ion Transport code System (PHITS) Monte Carlo transport code [26]. The simulated setup uses mono-energetic proton pencil beams between 100 MeV and 150 MeV incident on a homogeneous cylindrical water phantom with radius 15 cm and length 40 cm. No tissue heterogeneity, air cavity, or patient geometry is modelled.

The thesis starts from cone-level list-mode data. The input data are Hierarchical Data Format version 5 (HDF5) files generated with `ng-imager`, an existing NOVO preprocessing and imaging codebase developed by Hunter Ratliff [27]. These files contain fast-neutron cone parameters derived from the simulated interaction information. The SOE implementation developed here therefore does not model the full detector signal chain. Optical photon transport, photosensor response, finite timing and energy resolution, and event misclassificatio are outside the implemented scope. The thesis does not include clinical validation or dose reconstruction.

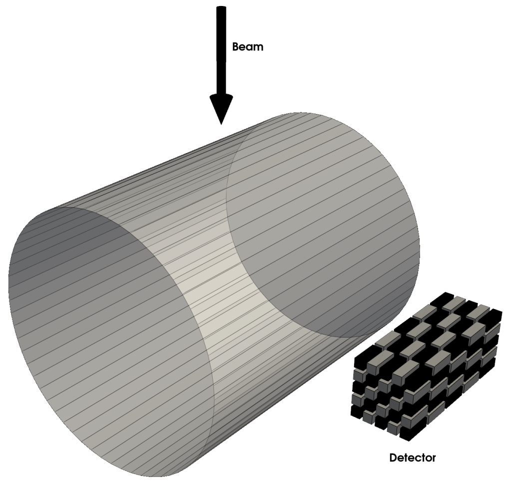


Figure 1.3: Simulation geometry used in the study. The segmented detector comprises 25 plastic-scintillator bars arranged in five layers of five, with alternating organic glass scintillator (OGS) and M600 materials. Each bar measures 1.2 cm × 1.2 cm × 14 cm. The beam travels in the $+z$ direction through a homogeneous cylindrical water phantom with radius 15 cm and length 40 cm. The phantom is aligned with the x-axis.

Although NOVO includes both prompt-gamma and fast-neutron detection, no prompt-gamma dataset was available for the present work.

The implementation is developed in Python as a documented, reproducible package with modular architecture. Performance-critical components are isolated so that compiled-language acceleration can be added later, but no compiled extension is used in the reported reconstructions. Computational performance is evaluated through measured execution time and scaling behaviour of the implemented pipeline. These measurements assess the practical cost of the tested reconstruction workflow and identify the gap to real-time use; they do not establish clinical real-time deployment.

Conclusions about reconstruction quality, range-surrogate behaviour, sampling behaviour, and computational performance apply only to the specific datasets, detector geometry, voxel grids, and SOE sampling regimes examined.

Unlike a development project with predefined product requirements, this thesis was formulated as a research-based reconstruction study. The written specification defined the work around three practical tasks: implementing a modular SOE reconstruction pipeline, evaluating reconstruction accuracy against simulated reference information, and assessing computational performance in terms of execution time, scalability, and memory use. Robustness under noise, data sparsity, and stochastic sampling conditions was included as an evaluation direction, while broader analyses such as uncertainty quantification, convergence studies, and cross-dataset generalisation were explicitly time permitting. Quasi-real-time feasibility was retained as a computational motivation, but it was deferred early in the project from a primary requirement to a secondary evaluation criterion because the immediate research question was whether SOE preserved range-relevant information in the available fast-neutron cone data.

1.5 Thesis outline

The report is organised to move from clinical motivation and project context to reconstruction theory, implementation, evaluation, and interpretation. Chapter 2 introduces proton range uncertainty, secondary-radiation range verification, fast-neutron detection, and the SOE reconstruction framework. Chapter 3 formalises the event-cone geometry, feasible reconstruction domain, proposal distribution, and Metropolis–Hastings target used by the reconstruction. Chapter 4 describes how the mathematical formulation is realised in the software pipeline, including event admission, sampling, diagnostics, and output reduction. Chapter 5 presents the reconstructed images, axial range-surrogate metrics, sampling diagnostics, and computational-performance measurements. Chapter 6 interprets the results in relation to the thesis objectives, with emphasis on range-surrogate behaviour, reconstruction limitations, and computational practicality. Chapter 7 summarises the main findings and identifies the conditions under which future work should extend the method.

Chapter 2

Background

2.1 Fast-neutron production as a range-surrogate signal

Range verification requires an observable that changes with the stopping depth of the primary proton beam [13, 16]. For fast-neutron measurements, the relevant link is the depth dependence of neutron production induced by primary protons in the irradiated medium [16].

Proton range is set by the cumulative energy lost by the primary protons as they traverse matter. Over therapeutic energies, the mean energy loss is dominated by electronic stopping, where Coulomb interactions transfer kinetic energy from the proton to atomic electrons. The corresponding stopping power depends on proton energy and absorber properties, including electron density and mean excitation potential [3]. In a material path, stopping depth is not determined by initial energy alone. It follows from the integrated stopping power along the beam path, often expressed through water-equivalent path length [3].

Fast neutrons arise from nuclear interactions. As primary protons traverse the irradiated medium, non-elastic interactions with target nuclei can eject secondary neutrons [16]. The local neutron-production rate depends on the residual proton fluence, the proton energy at that depth, and the relevant nuclear reaction cross sections [28]. Because the primary beam loses energy and particles along its path, the fast-neutron production field changes with depth.

The distal falloff of that production field is the feature that connects fast neutrons to proton range. Primary-proton-induced neutron production requires primary protons with sufficient residual energy to undergo nuclear interactions. Near the end of the proton track, both surviving primary fluence and residual energy decrease, so neutron production falls as the beam approaches its stopping region [16]. The distal falloff of the fast-neutron production profile can therefore be tested as a surrogate observable for proton stopping depth [16].

The fast-neutron production profile should not be read as a dose profile. Dose deposition is governed mainly by electronic stopping and increases toward the Bragg peak [3]. Fast-neutron production is governed by nuclear interaction probability and surviving primary fluence [16]. Its maximum need not coincide with the Bragg peak. For range verification, the relevant information is the depth dependence of the production field, especially its distal falloff, rather than the production maximum itself [16].

Neutron transport separates the production field from the detected event distribution.

After production, fast neutrons can scatter in the target before reaching the detector, changing their direction and energy [16]. The detector therefore does not observe the production profile directly. It records interaction histories from which possible emission origins must be reconstructed [17].

2.2 Double-scatter events and cone constraints

A double-scatter event records a fast neutron that interacts in two distinct detector elements. The first interaction transfers energy to a recoil proton in the scintillator. The displacement between the two interaction sites gives the scattered-neutron direction in the ideal event model. Together with the time separation between the interactions, it also determines the scattered-neutron kinetic energy [29]. The two recorded interaction sites therefore provide directional information, but they do not identify the emission point.

The cone constraint follows from elastic-scattering kinematics. Let E_r denote the recoil-proton kinetic energy deposited at the first scatter, and let E'_n denote the kinetic energy of the neutron after that scatter. Conservation of kinetic energy gives the incident neutron energy,

$$E_n = E_r + E'_n.$$

For non-relativistic neutron–proton elastic scattering, the scattering angle θ between the incident and scattered neutron directions satisfies

$$\theta = \arctan\sqrt{\frac{E_r}{E'_n}}.$$

The measured recoil-proton kinetic energy and scattered-neutron energy therefore determine only the polar angle of the incoming trajectory, not its azimuth [29]. That unresolved azimuth defines a cone of possible incident neutron directions. Back-projected from the first interaction site, the directional cone becomes a conical surface of possible source locations.

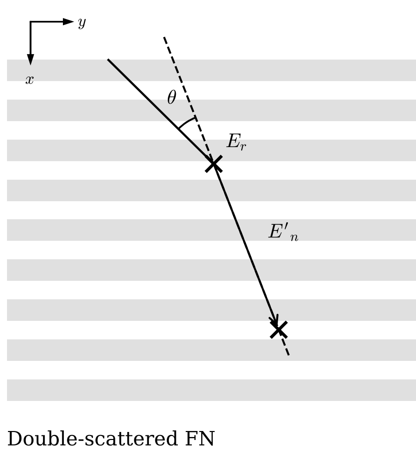


Figure 2.1: Double-scatter fast-neutron event and the resulting cone constraint.

In the ideal event model, the first interaction site gives the cone apex. The measured scattered-neutron direction fixes the cone axis. The scattering angle gives the cone opening angle. Each possible emission point for the event lies on that conical surface [29].

Finite energy resolution broadens the inferred values of E_r and E_n'. Finite position and timing resolution perturb the cone axis and the scattered-neutron energy. Interactions on carbon or other detector nuclei can violate the neutron–proton recoil model. Neutron scattering before detection can also make the detected direction differ from the original emission direction [30]. After event construction and selection, the resulting simulated events are treated as exact cone constraints.

2.3 Stochastic Origin Ensemble reconstruction

A single cone does not determine a neutron emission origin; it defines a continuum of geometrically possible origins. Many events therefore give many conical surfaces. The reconstruction task is to estimate an emission-origin density from the regions where these cone constraints are mutually consistent.

SOE reconstruction represents this inverse problem by assigning one tentative origin to each detected event. The collection of tentative origins forms the current reconstruction configuration. When these origins are accumulated on a voxel grid, the resulting occupancies provide an image-like estimate of the source distribution. The image is therefore not obtained from one event at a time. It emerges from the joint configuration of all event origins.

The SOE update mechanism changes that joint configuration iteratively. For a selected event, the algorithm proposes a new origin on that event's cone. The proposal is accepted or rejected by a sampling rule that favours configurations with stronger local support from the ensemble. Origins are therefore redistributed toward regions supported by several event cones, while isolated or weakly supported placements are suppressed. Repeated updates turn the cone ensemble into a sampled estimate of the underlying emission distribution.

This places the present reconstruction in the SOE lineage. Sitek first introduced SOE for emission tomography under incomplete event-origin information [18]. Andreyev et al. extended the method to cone-based reconstruction, where each event restricts the origin to a conical surface [20]. Mackin et al. then used cone-based SOE for prompt-gamma Compton imaging in proton therapy [21].

The thesis makes the corresponding substitution for fast-neutron range verification: prompt-gamma Compton cones are replaced by fast-neutron double-scatter cones, while the reconstruction remains an ensemble sampler over event-specific conical surfaces. Chapter 3 turns this reconstruction principle into a sampling problem.

Chapter 3

Theoretical Formulation

The sampling problem is defined by four objects: the volume of interest (VOI), the voxel grid, the retained events, and the feasible cone surface associated with each event. Together these objects define the admissible SOE state space, where one representative origin is assigned to each event. The target distribution weights admissible states through their induced voxel occupancies. A single-event Metropolis–Hastings kernel then samples this target by updating one event origin at a time [18, 20, 31, 32].

3.1 Geometric problem setup

The geometric setup fixes the domain in which representative origins may lie and the grid on which they are later counted. The VOI and voxelisation define the continuous-to-discrete map. The event model then restricts each loaded event to the part of its canonical cone surface lying inside the VOI.

3.1.1 VOI and voxelisation

The reconstruction is carried out on a fixed three-dimensional VOI. All spatial quantities are expressed in a common Cartesian coordinate system. Let

$$\Omega := [x_{\min}, x_{\max}) \times [y_{\min}, y_{\max}) \times [z_{\min}, z_{\max}) \subset \mathbb{R}^3$$

denote the VOI.

To obtain a discrete image representation, the VOI is partitioned into a voxel grid with n_x, n_y, and n_z voxels along the x-, y-, and z-axes, respectively. The corresponding voxel spacings are

$$\Delta x = \frac{x_{\max} - x_{\min}}{n_x}, \qquad \Delta y = \frac{y_{\max} - y_{\min}}{n_y}, \qquad \Delta z = \frac{z_{\max} - z_{\min}}{n_z}.$$

Let

$$I_x := \{0, \ldots, n_x - 1\}, \qquad I_y := \{0, \ldots, n_y - 1\}, \qquad I_z := \{0, \ldots, n_z - 1\},$$

and define the voxel index set by $V := I_x \times I_y \times I_z$. For each $(i, j, k) \in V$, the corresponding voxel cell is

$$\begin{aligned} B_{i,j,k} := [&x_{\min} + i\Delta x,\ x_{\min} + (i+1)\Delta x) \\ \times [&y_{\min} + j\Delta y,\ y_{\min} + (j+1)\Delta y) \\ \times [&z_{\min} + k\Delta z,\ z_{\min} + (k+1)\Delta z). \end{aligned}$$

The half-open convention makes the cells disjoint while preserving full coverage of Ω. Points on shared voxel faces are therefore assigned to exactly one voxel.

The centre of voxel (i, j, k) is

$$c_{i,j,k} = \big(x_{\min} + (i + \tfrac{1}{2})\Delta x,\ y_{\min} + (j + \tfrac{1}{2})\Delta y,\ z_{\min} + (k + \tfrac{1}{2})\Delta z\big).$$

The world-to-voxel map links the continuous reconstruction geometry to the discrete image representation. Define

$$\nu : \mathbb{R}^3 \to V \cup \{\perp\}$$

by

$$\nu(x, y, z) = \begin{cases} \left(\left\lfloor \dfrac{x - x_{\min}}{\Delta x} \right\rfloor, \left\lfloor \dfrac{y - y_{\min}}{\Delta y} \right\rfloor, \left\lfloor \dfrac{z - z_{\min}}{\Delta z} \right\rfloor \right), & (x, y, z) \in \Omega, \\ \perp, & (x, y, z) \notin \Omega. \end{cases}$$

Because the cells $B_{i,j,k}$ partition Ω, the map $\nu(r)$ returns the unique voxel index for each point $r \in \Omega$, and $\perp$ otherwise. This continuous-to-discrete assignment is the bookkeeping step that connects the geometric event model to the occupancy image used by the SOE target distribution, as shown schematically in fig. 3.1.

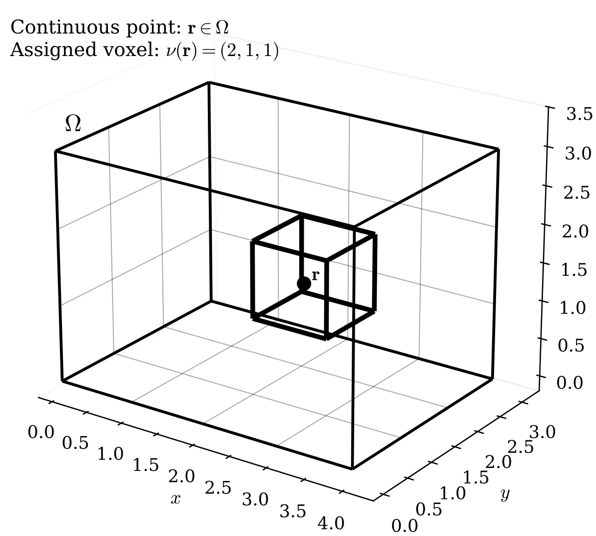


Figure 3.1: Continuous-to-discrete assignment in the VOI. The point $r \in \Omega$ is assigned by ν to the unique half-open voxel cell containing it.

3.1.2 Event model, cone geometry, and feasible surfaces

Let E_0 denote the finite index set of all loaded events. For each $e \in E_0$, define the canonical event datum

$$d_e := (a_e, u_e, \theta_e),$$

following the cone-event representation used for double-scatter neutron imaging [33], where $a_e \in \mathbb{R}^3$ is the cone apex, $u_e \in \mathbb{S}^2$ is a unit axis direction on the unit sphere, and $\theta_e \in (0, \pi/2)$ is the cone half-opening angle. The axis is interpreted using a fixed convention, so that u_e always specifies the forward direction of the retained single nappe; this removes the orientation ambiguity of the underlying double cone. Thus each event is represented by a canonical half-cone.

Given d_e, let

$$\lambda_e := \cos\theta_e \in (0, 1).$$

The corresponding single-nappe cone surface is defined by

$$C_e := \left\{ r \in \mathbb{R}^3 \setminus \{a_e\} : \frac{(r - a_e) \cdot u_e}{\|r - a_e\|} = \lambda_e \right\}.$$

The apex is excluded because the defining expression is undefined at $r = a_e$. This defines the event-wise cone surface before restriction to the reconstruction domain.

The feasible surface of event e is the portion of its cone that lies inside the reconstruction domain [20, 21],

$$\Sigma_e := C_e \cap \Omega.$$

The set Σ_e is treated as a two-dimensional surface equipped with its induced surface-area measure, denoted by μ_e; the formal Hausdorff-measure definition and the relevant finiteness properties are given in appendix A.1. An event is called usable if and only if its feasible surface has strictly positive surface area, so that the event admits a nondegenerate admissible surface and the corresponding surface-area law is well-defined. The surviving-event set is therefore

$$E := \{e \in E_0 : \Sigma_e \text{ has strictly positive surface area}\}.$$

Since Σ_e is generally a surface rather than a singleton, each event contributes an event-wise feasible surface within the reconstruction domain. These feasible surfaces provide the geometric basis for the state-space formulation.

3.2 State space and image representation

The unknown is the event-indexed collection of representative origins, where each representative origin is constrained to lie on the feasible surface of its own surviving event [18, 20, 21].

Write

$$N := |E|.$$

For each $e \in E$ the SOE state is the vector

$$\mathbf{s} := (x_e)_{e \in E}, \qquad x_e \in \mathbb{R}^3,$$

where x_e is the representative origin assigned to event e.

A state $\mathbf{s}$ is called *admissible* if and only if every representative origin lies on its corresponding feasible surface, that is,

$$x_e \in \Sigma_e \qquad \forall\, e \in E.$$

Accordingly, the admissible state space is the Cartesian product

$$\mathcal{S} := \prod_{e \in E} \Sigma_e.$$

Hence $\mathcal{S}$ is an event-wise constrained state space: each component is restricted by the feasible geometry of its own event. The voxel grid image is not part of the Markov state itself; it is a deterministic image derived from the continuous representative-point state $\mathbf{s}$ [18, 20].

The image induced by a state $\mathbf{s} \in \mathcal{S}$ is obtained by voxelising the representative origins through the world-to-voxel map. Writing B_v for the voxel cell indexed by $v \in V$, that is, $B_v = B_{i,j,k}$ when $v = (i,j,k)$, define for each voxel $v \in V$ the occupancy

$$n_v(\mathbf{s}) := \sum_{e \in E} \mathbf{1}_{\{\nu(x_e)=v\}},$$

where $\mathbf{1}_{\{\nu(x_e)=v\}}$ is the indicator that the representative origin of event e lies in voxel v. The discrete image induced by $\mathbf{s}$ is therefore the voxelwise occupancy field

$$\mathbf{n}(\mathbf{s}) := \big(n_v(\mathbf{s})\big)_{v \in V}.$$

For reporting, one may also define the corresponding volume-rescaled density image

$$\rho_v(\mathbf{s}) := \frac{n_v(\mathbf{s})}{|B_v|}.$$

Because the voxel grid is Cartesian with fixed spacings $\Delta x, \Delta y, \Delta z$, every voxel has volume

$$|B_v| = \Delta x\, \Delta y\, \Delta z \qquad \forall v \in V.$$

The occupancy image $\mathbf{n}(\mathbf{s})$ is the canonical image used in the target law and acceptance mechanism later on [18, 20], because those constructions are written in terms of voxel counts; $\rho_v(\mathbf{s})$ is only the corresponding volume-rescaled output quantity.

A basic invariant follows from the occupancy definition. Since each admissible representative origin is assigned by the world-to-voxel map to exactly one voxel of the reconstruction grid, one has

$$\sum_{v \in V} n_v(\mathbf{s}) = N \qquad \forall \mathbf{s} \in \mathcal{S}.$$

This identity expresses conservation of event count: the image redistributes the N representative origins over voxels, but does not create or remove them.

3.3 Target distribution and modelling principle

While the state space $\mathcal{S}$ determines which SOE configurations are admissible, the target distribution determines how those admissible states are weighted. As in the literature, this weighting depends on a state through the induced voxel occupancy image rather than through the continuous locations themselves [18, 20]. This choice will later make the Metropolis–Hastings ratio depend only on the voxels affected by a proposed single-event move. Thus two admissible states with the same voxel occupancies receive the same target weight even if their representative origins are arranged differently within the voxels.

Because each admissible state is a collection of event origins constrained to event-specific feasible surfaces, the target density is not defined relative to volume measure, but relative to the product admissible-surface measure

$$\mu := \bigotimes_{e\in E} \mu_e$$

on $\mathcal{S} = \prod_{e\in E} \Sigma_e$. In practical terms, μ is the reference measure obtained by combining the per-event surface-area measures across all event components.

To obtain a tractable occupancy-based target, introduce auxiliary voxel weights $p = (p_v)_{v\in V}$ and condition on the voxel assignments induced by a state $\mathbf{s}$. Given p, the occupancy-only likelihood kernel is proportional to $\prod_{v\in V} p_v^{n_v(\mathbf{s})}$. Equipping p with a symmetric Dirichlet prior with parameter $\alpha > 0$ [34] and marginalising over p yields a gamma-product form in the occupancies (see appendix A.2). The auxiliary weights are therefore not additional SOE variables; they are integrated out to define a state weight that depends only on the occupancy counts.

For a parameter $\alpha > 0$, the target is then taken to have density

$$\pi_\alpha(\mathbf{s}) = \frac{1}{Z_\alpha} \prod_{v\in V} \Gamma\big(n_v(\mathbf{s}) + \alpha\big), \qquad \mathbf{s} \in \mathcal{S}, \tag{3.1}$$

with respect to μ, where Γ denotes the gamma function, which extends the factorial to positive real arguments, and Z_α denotes the normalising constant. In particular, $\Gamma(k+1) = k!$ for integers $k \geq 0$. The gamma-function recurrence is

$$\Gamma(z + 1) = z\,\Gamma(z).$$

The remaining Dirichlet normalising factors are independent of $\mathbf{s}$ and are therefore absorbed into Z_α in (3.1). Appendix A.2 gives the marginalisation and proves that $0 < Z_\alpha < \infty$.

For $\alpha = 1$,

$$\pi_1(\mathbf{s}) \propto \prod_{v\in V} n_v(\mathbf{s})!.$$

Since $\pi_\alpha(\mathbf{s})$ depends on $\mathbf{s}$ only through the occupancies $n_v(\mathbf{s})$, all admissible states with the same voxel-count image receive the same target weight. For $\alpha = 1$ and the total count N, this approximation assigns larger weight to some more uneven voxel-count allocations than to more evenly split ones. For example, if counts a and b in two voxels are merged into a single voxel, then

$$(a + b)! \geq a!\,b!,$$

so the factorial product does not decrease under such a merge.

3.4 Proposal mechanism

The proposal mechanism specifies how candidate moves between admissible states are generated before the acceptance rule is applied. Each proposal changes exactly one event-component and leaves all other components fixed [18, 20]. Here M denotes the randomly selected event index, while m denotes a realised value.

An event index $M \in E$ is selected uniformly at random:

$$\mathbb{P}(M = m) = \frac{1}{N}, \qquad m \in E. \tag{3.2}$$

For a realised index $m \in E$, the proposal redraws only that event's representative origin. The new origin is drawn on the feasible surface Σ_m. In the formulation, this random cone-surface draw is specified as uniform with respect to the induced surface-area measure, formalising the random cone-surface proposal used in SOE [20, 21].

$$Q_m(A) := \frac{\mu_m(A \cap \Sigma_m)}{\mu_m(\Sigma_m)}, \qquad A \subseteq \mathbb{R}^3 \text{ Borel.} \tag{3.3}$$

Because $m \in E$ is a surviving event, $\mu_m(\Sigma_m) > 0$ by the event-usability criterion. The bound $\mu_m(\Sigma_m) < \infty$ is established in appendix A.1 under the surface-measure construction. Hence (3.3) defines a probability law on Σ_m.

If x'_m denotes a random draw from Q_m, then $x'_m \in \Sigma_m$ almost surely. Equivalently, with respect to μ_m, the within-event proposal density is

$$q_m(x'_m \mid x_m) = \frac{1}{\mu_m(\Sigma_m)}, \qquad x_m, x'_m \in \Sigma_m, \tag{3.4}$$

and is zero outside Σ_m. In particular, the within-event proposal is independent of the current location x_m.

The candidate state is formed by replacing only the selected component:

$$x'_e = \begin{cases} x'_m, & e = m, \\ x_e, & e \neq m. \end{cases} \tag{3.5}$$

Equivalently, the candidate state is

$$\mathbf{s}' := (x'_e)_{e \in E}. \tag{3.6}$$

Since $x'_m \in \Sigma_m$ almost surely and all remaining components are unchanged, one has $\mathbf{s}' \in \mathcal{S}$. The proposal therefore preserves admissibility and changes only the selected event's voxel assignment, which is why the acceptance rule can be expressed locally. The single-event structure of the proposal is shown schematically in fig. 3.2.

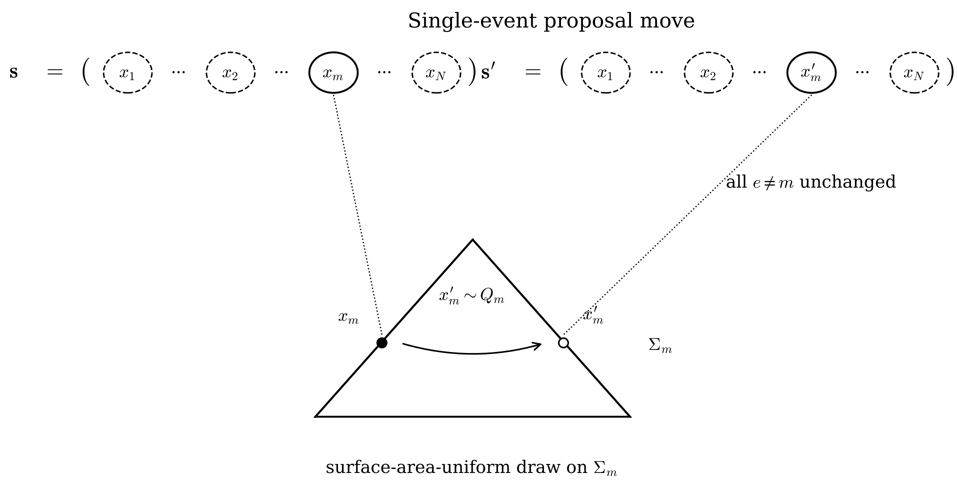


Figure 3.2: Single-event proposal move. The selected component x_m is redrawn as $x'_m \sim Q_m$ on the same feasible surface Σ_m, while all components with $e \neq m$ are left unchanged.

3.5 Acceptance logic

The acceptance rule corrects the candidate moves so that the Markov chain targets π_α. For a single-event proposal updating event m from x_m to x'_m, the Metropolis–Hastings acceptance probability is [31, 32]

$$a(\mathbf{s}, \mathbf{s}') := \min\left\{1,\ \frac{\pi_\alpha(\mathbf{s}')}{\pi_\alpha(\mathbf{s})} \frac{q_m(x_m \mid x'_m)}{q_m(x'_m \mid x_m)}\right\}. \tag{3.7}$$

Write

$$v^- := \nu(x_m), \qquad v^+ := \nu(x'_m).$$

For every voxel $v \in V$, the single-event move changes the occupancy by

$$n_v(\mathbf{s}') = n_v(\mathbf{s}) - \mathbf{1}_{\{v=v^-\}} + \mathbf{1}_{\{v=v^+\}}.$$

Thus, if $v^- = v^+$, no voxel occupancy changes. If $v^- \neq v^+$, only two voxels change: v^- loses one count and v^+ gains one count. This locality is shown schematically in fig. 3.3.

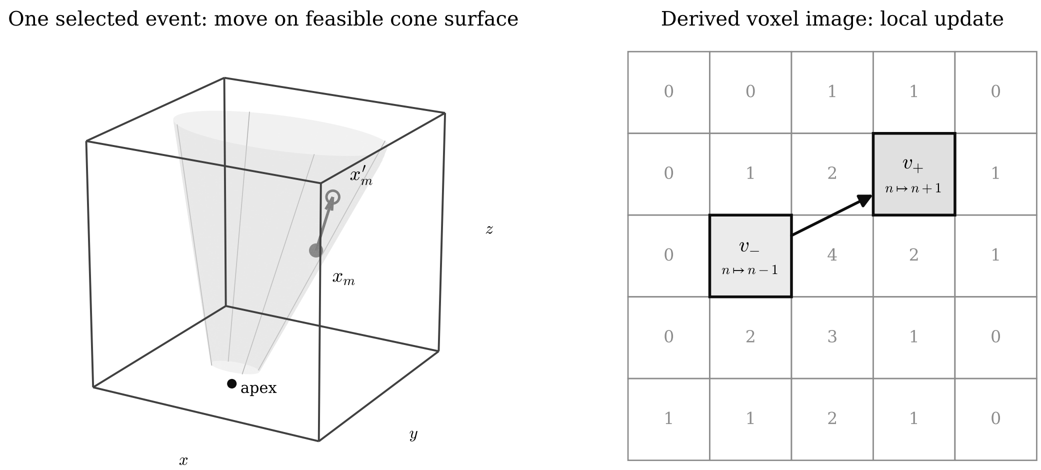


Figure 3.3: Local voxel-image update induced by a single-event proposal. The selected representative origin moves from x_m to x'_m on its feasible cone surface. In the derived voxel image, the departure voxel v^- loses one count and the arrival voxel v^+ gains one count, while all other voxel occupancies remain unchanged.

Substituting (3.1) into the target ratio, all factors corresponding to unchanged voxels cancel. One obtains

$$\frac{\pi_\alpha(\mathbf{s}')}{\pi_\alpha(\mathbf{s})} = \begin{cases} 1, & v^+ = v^-, \\ \dfrac{\Gamma\big(n_{v^-}(\mathbf{s}) - 1 + \alpha\big)\,\Gamma\big(n_{v^+}(\mathbf{s}) + 1 + \alpha\big)}{\Gamma\big(n_{v^-}(\mathbf{s}) + \alpha\big)\,\Gamma\big(n_{v^+}(\mathbf{s}) + \alpha\big)}, & v^+ \neq v^-. \end{cases} \tag{3.8}$$

Since the updated event currently occupies voxel v^-, one has

$$n_{v^-}(\mathbf{s}) \geq 1.$$

Using the gamma-function recurrence $\Gamma(z+1) = z\,\Gamma(z)$, this simplifies to

$$\frac{\pi_\alpha(\mathbf{s}')}{\pi_\alpha(\mathbf{s})} = \begin{cases} 1, & v^+ = v^-, \\ \dfrac{n_{v^+}(\mathbf{s}) + \alpha}{n_{v^-}(\mathbf{s}) - 1 + \alpha}, & v^+ \neq v^-. \end{cases} \tag{3.9}$$

For the uniform surface-area proposal, the within-event proposal is symmetric, so

$$q_m(x_m \mid x'_m) = q_m(x'_m \mid x_m),$$

and the proposal ratio in (3.7) is therefore equal to one. The acceptance probability then becomes

$$a(\mathbf{s}, \mathbf{s}') = \begin{cases} 1, & v^+ = v^-, \\ \min\Big\{1, \frac{n_{v^+}(\mathbf{s}) + \alpha}{n_{v^-}(\mathbf{s}) - 1 + \alpha}\Big\}, & v^+ \neq v^-. \end{cases} \tag{3.10}$$

Setting $\alpha = 1$, (3.10) reduces to

$$a(\mathbf{s}, \mathbf{s}') = \begin{cases} 1, & v^+ = v^-, \\ \min\left\{1, \frac{n_{v^+}(\mathbf{s})+1}{n_{v^-}(\mathbf{s})}\right\}, & v^+ \neq v^-. \end{cases} \tag{3.11}$$

This has the same occupancy-ratio form used in SOE litterature [18, 20], but here it follows from (3.1) and the symmetric surface-area proposal.

Under the target π_α and the symmetric surface-area proposal, the resulting Metropolis–Hastings kernel satisfies detailed balance with respect to π_α [35]; a proof is deferred to appendix A.3. In particular, π_α is invariant under this kernel.

3.6 MCMC execution and output

A run is specified by an admissible initial state, a total number of steps, a burn-in length, a thinning interval, and an output estimator, following standard Markov chain Monte Carlo (MCMC) reporting conventions [35].

The chain is started from an admissible initial state

$$\mathbf{s}^{(0)} \in \mathcal{S}.$$

No particular distribution is required for $\mathbf{s}^{(0)}$. The only requirement is event-wise admissibility,

$$x_e^{(0)} \in \Sigma_e, \qquad e \in E.$$

A *step* is one single-event proposal together with its accept-or-reject decision. A *sweep* is an accounting unit equal to N successive steps, where N is the number of surviving events. This follows the random single-event update convention used in SOE [18, 20]. Since event indices are drawn randomly, one sweep does not mean that each event is selected exactly once.

Write

$$\mathbf{s}^{(t)}, \qquad t = 0, 1, \ldots, K,$$

for the state after t completed steps, where K is the total number of steps executed. Let $B \in \mathbb{N}_0$ denote the burn-in length in steps. The first B steps are discarded as burn-in, so that the reported output is not taken from the initial transient part of the chain.

After burn-in, states are retained at a fixed step interval $T \in \mathbb{N}$. The retained step indices are

$$t_c := B + cT, \qquad c = 1, \ldots, C,$$

with $t_C \leq K$. Equivalently,

$$C = \left\lfloor \frac{K - B}{T} \right\rfloor.$$

The run parameters must satisfy $C \geq 1$, so that at least one post-burn-in state is retained and the reported average is well defined. If no thinning is used, then $T = 1$, so every post-burn-in step is retained.

For each retained state $\mathbf{s}^{(t_c)}$, the induced image is the occupancy field

$$\mathbf{n}\big(\mathbf{s}^{(t_c)}\big) = \big(n_v(\mathbf{s}^{(t_c)})\big)_{v\in V}.$$

The primary reported reconstruction is the voxel-wise mean occupancy over the retained states:

$$\bar{n}_v := \frac{1}{C}\sum_{c=1}^{C} n_v\big(\mathbf{s}^{(t_c)}\big), \qquad v \in V. \tag{3.12}$$

Under the target π_α, this retained-state average is interpreted as the SOE ensemble-average image estimator [18], with the retained post-burn-in states treated as approximately stationary MCMC samples [35]. Since each retained state satisfies $\sum_{v\in V} n_v(\mathbf{s}^{(t_c)}) = N$, the averaged output also satisfies

$$\sum_{v\in V} \bar{n}_v = N.$$

If the reconstruction is reported in density form, the corresponding output is obtained by the same voxel-wise rescaling used for individual states:

$$\bar{\rho}_v := \frac{1}{C}\sum_{c=1}^{C} \rho_v\big(\mathbf{s}^{(t_c)}\big) = \frac{\bar{n}_v}{|B_v|}, \qquad v \in V. \tag{3.13}$$

Thus the reported image may be expressed either as mean occupancy or as mean density.

A terminal-state image may also be reported as a secondary output:

$$\mathbf{n}\big(\mathbf{s}^{(K)}\big),$$

or, equivalently, its density version $\rho_v(\mathbf{s}^{(K)})$. The main reported reconstruction remains the retained-state average in (3.12), or its density form in (3.13).

Chapter 4

Computational Realisation

Chapter 3 defined the reconstruction model as an exact mathematical object. The computational task is to realise that formulation with finite data structures, floating-point geometry, and persisted reconstruction outputs.

This chapter describes where that realisation preserves the mathematical contract and where controlled numerical procedures enter. Input loading constructs runtime events from the upstream cone data. Finite geometry and event admission replace exact continuous predicates with validated floating-point tests and deterministic witnesses. The reconstruction kernel then implements the single-event proposal, acceptance rule, retained-state accumulation, and diagnostic sidecars. The final offline reductions keep reconstruction separate from downstream evaluation by defining how persisted outputs, PHITS truth files, and geometry-only diagnostics are converted into the quantities reported in chapter 5.

4.1 Input loading and event construction

The implementation begins with two external inputs: a job configuration and an upstream HDF5 file. The configuration fixes the VOI, voxel counts, registration transform, stochastic seeds, run controls, artifact destinations, and the species filter. The upstream file provides the per-event cone rows from which events are constructed.

Geometric event data are read from the `/cones` group of the upstream file [27]. The accepted datasets are `apex_xyz` for the raw apex $a_{\mathrm{raw}} \in \mathbb{R}^3$, `axis_xyz` for the raw axis $u_{\mathrm{raw}} \in \mathbb{R}^3$, and `theta_rad` for the raw half-angle $\theta_{\mathrm{raw}} \in \mathbb{R}$. A companion `/cones/species` dataset carries per-event particle labels. From the dataset, the implementation decodes numeric species codes to strings. The species filter then discards every non-neutron row, so that only neutron events proceed to construction.

Construction maps each surviving row into the reconstruction frame and converts it into the runtime event representation. Before any row is processed, the rigid transform is validated: the rotation matrix Q must satisfy $\|Q^\top Q - I\|_{\max} < 10^{-10}$ and $|\det Q - 1| < 10^{-10}$, and a translation vector g must be finite. Each retained row is then registered by

$$a = Q\, a_{\mathrm{raw}} + g, \qquad u = Q\, u_{\mathrm{raw}},$$

and the registered axis is normalised. Rows with a non-finite apex, axis, half-angle, or with post-registration norm $\|u\| \leq 10^{-12}$, are discarded.

The final step enforces orientation. When $\cos\theta_{\text{raw}} < 0$, the axis and half-angle are replaced by

$$u \leftarrow -u, \qquad \theta \leftarrow \pi - \theta_{\text{raw}},$$

which maps the datum to the forward nappe without changing the underlying cone. Rows with $\theta = 0$ or $\theta = \pi/2$ are discarded as degenerate. Every surviving event therefore satisfies $\theta \in (0, \pi/2)$. After this filtering stage, the retained-row index set is E_0.

4.2 Geometry and event admission

The mathematical formulation in chapter 3 defines the world-to-voxel map, feasible surfaces, and admissible representative points as exact objects. The implementation must convert these objects into a finite event set and a valid initial state under floating-point arithmetic. The world-to-voxel map can disagree with the VOI containment check at cell boundaries; the exact event-usability condition $\mu_e(\Sigma_e) > 0$ is certified through a finite azimuth-sampled witness test; and the initial representative point must be constructed and validated through several independent finite-precision stages, each of which can fail.

4.2.1 Voxelisation policy

The world-to-voxel map assigns each point in the VOI a unique voxel index and returns $\perp$ for every point outside. In exact arithmetic, the half-open convention prevents a coordinate satisfying $x_{\min} \leq x < x_{\max}$ from producing the index n_x. Under floating-point arithmetic, however, the VOI containment predicate and the normalised coordinate used in the floor operation are evaluated separately. A coordinate accepted by the containment check can therefore still produce

$$\left\lfloor \frac{x - x_{\min}}{\Delta x} \right\rfloor = n_x,$$

and analogously on the other axes. The containment check and the discrete index computation then disagree.

Clamping an out-of-range index to the nearest valid value would force a boundary point into a neighbouring voxel, introducing a deterministic bias into the occupancy field. The implementation instead rejects: after the VOI containment check and the coordinate-wise floor computation, the candidate index is accepted only if

$$0 \leq i < n_x, \qquad 0 \leq j < n_y, \qquad 0 \leq k < n_z.$$

If any component fails, the map returns $\perp$, represented at runtime by `None`. Every accepted point therefore maps to one valid half-open voxel cell.

4.2.2 Sampled admission gate

The SOE literature specifies where representative origins are allowed to lie, but it does not define a separate preprocessing test for admitting or discarding events [18, 20, 21].

In the formulation of section 3.1, the requirement is made explicit by defining a usable event as one whose feasible cone surface inside the VOI is nondegenerate. This subsection describes the finite implementation of that exact condition. The implemented gate is a one-sided preprocessing approximation to the exact usability condition. It is used only to certify a finite event pool before MCMC execution, so the implementation seeks positive interior witnesses rather than solving the full continuous cone–box admission problem exactly.

For $e \in E_0$ the implementation attaches a deterministic orthonormal frame (p_e, q_e, u_e) to the event axis. For each azimuth $\xi \in [0, 2\pi)$, the corresponding unit generator direction is

$$w_e(\xi) = \lambda_e \, u_e + \sqrt{1 - \lambda_e^2}\big(\cos\xi \, p_e + \sin\xi \, q_e\big), \tag{4.1}$$

and the associated ray is

$$r_{e,\xi}(\ell) := a_e + \ell \, w_e(\xi), \qquad \ell > 0.$$

Write

$$\Omega^\circ := (x_{\min}, x_{\max}) \times (y_{\min}, y_{\max}) \times (z_{\min}, z_{\max})$$

for the interior of the VOI. For a nondegenerate cone in this cone–box setting, positive feasible surface measure is equivalent to the existence of at least one azimuth whose generator ray enters Ω°. Indeed, an interior entry point gives, by openness of Ω° and continuity of the cone parametrisation, a local surface patch of positive measure. Conversely, a positive-measure feasible surface cannot be contained entirely in the boundary of the box. The implementation tests that condition on a deterministic midpoint grid in azimuth,

$$\Xi^{(K_{\mathrm{az}})} := \left\{ \xi_m = \frac{2\pi}{K_{\mathrm{az}}}\big(m + \tfrac{1}{2}\big) : m = 0, \ldots, K_{\mathrm{az}} - 1 \right\}.$$

For each sampled azimuth ξ, the open radial support

$$I_e^\circ(\xi) := \big\{ \ell > 0 : r_{e,\xi}(\ell) \in \Omega^\circ \big\}$$

is obtained by coordinate-wise box–ray interval intersection; the endpoint formulas are given in appendix A.4. The admission decision is

$$\widehat{U}_e^{K_{\mathrm{az}}} := \mathbf{1}\big\{ \exists\, \xi \in \Xi^{(K_{\mathrm{az}})} : I_e^\circ(\xi) \neq \varnothing \big\}. \tag{4.2}$$

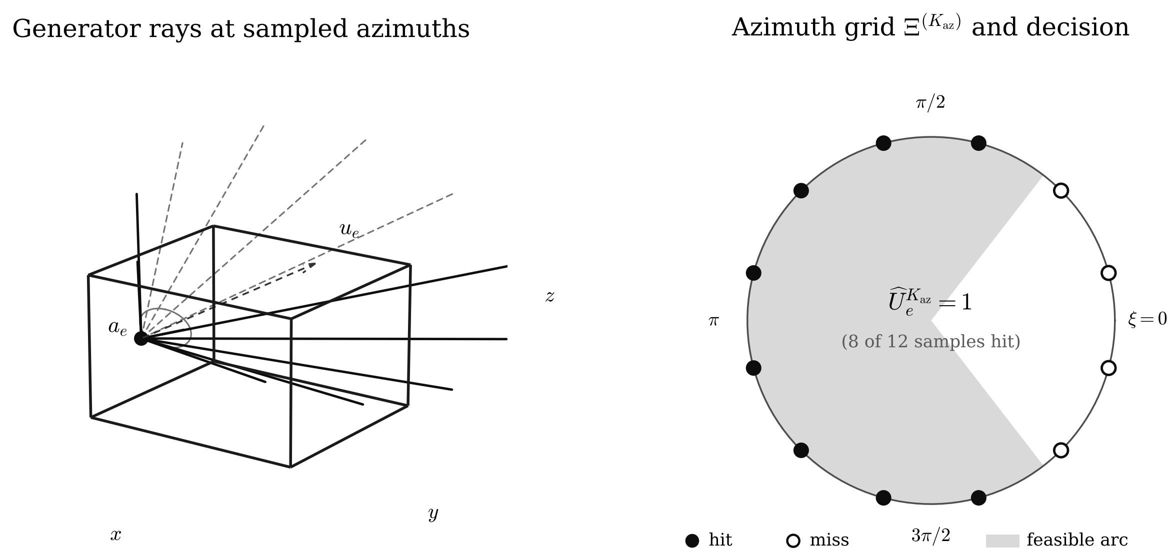


Figure 4.1: Sampled admission gate for cone–box usability. The left panel shows generator rays obtained from a finite azimuth grid on the cone surface. The right panel illustrates the corresponding azimuth-domain decision for an example with $K_{\mathrm{az}} = 12$: filled markers denote sampled azimuths whose generator ray enters Ω°, while open markers denote missed samples. The shaded feasible arc is shown only for illustration; the implemented gate uses the sampled witnesses, not the continuous arc itself.

Figure 4.1 illustrates the finite-grid nature of the implemented admission test and the distinction between sampled witnesses and the underlying continuous usability condition.

The gate is one-sided. In exact arithmetic, every admitted event is usable, but some usable events may be missed. If $\widehat{U}_e^{K_{\mathrm{az}}} = 1$, the witness generator enters Ω°, so usability holds under the exact criterion. If $\widehat{U}_e^{K_{\mathrm{az}}} = 0$, no sampled azimuth witnesses interior entry, but an interior-entering generator may still exist between sample points. The certified event set is

$$\widehat{E}^{\mathrm{gate}}_{K_{\mathrm{az}}} := \left\{ e \in E_0 : \widehat{U}_e^{K_{\mathrm{az}}} = 1 \right\}.$$

Under the exact strict-interior witness interpretation,

$$\widehat{E}^{\mathrm{gate}}_{K_{\mathrm{az}}} \subseteq E.$$

Thus the sampled gate can introduce false negatives but not false positives, apart from ordinary finite-precision failures handled by the later initialisation validation.

The sampling density K_{az} is configurable. The reported runs use $K_{\mathrm{az}} = 512$, so each event is tested by a fixed number of box–ray intersections. The resulting preprocessing cost is $O(K_{\mathrm{az}})$ per event. By contrast, exact continuous admission would require constructing the event-specific cone–box azimuth partition before deciding whether any feasible segment has positive surface measure.

4.2.3 Deterministic initial representative points

Each event in $\widehat{E}^{\text{gate}}_{K_{\text{az}}}$ has passed the sampled admission test but does not yet hold a concrete representative point. In the SOE literature, the initial representative points are chosen randomly [18, 20, 21]. The implementation instead uses a deterministic initialiser. This is not a change to the SOE target distribution or proposal law; it only fixes one admissible starting configuration. As in the literature, the starting point may affect the time needed to approach equilibrium. Provided the chain mixes on the realised state space, post-burn-in samples are governed by the Markov kernel rather than by the particular initialiser.

The initialisation step therefore turns the sampled interior witness into one admissible representative point and then validates that point against the voxelisation policy. The deterministic rule below is used only for the initial state, not for the subsequent proposal moves.

Let $e \in \widehat{E}^{\text{gate}}_{K_{\text{az}}}$. The initialiser scans the midpoint azimuth grid $\Xi^{(K_{\text{az}})}$ in deterministic order and selects the first azimuth ξ^*_e whose strict-open radial support is nonempty,

$$I^\circ_e(\xi^*_e) \neq \varnothing.$$

Writing

$$I^\circ_e(\xi^*_e) = \left(\ell^{\circ,-}_e(\xi^*_e),\ \ell^{\circ,+}_e(\xi^*_e)\right),$$

the initialiser places the representative point at the midpoint radius

$$\ell^*_e := \frac{\ell^{\circ,-}_e(\xi^*_e) + \ell^{\circ,+}_e(\xi^*_e)}{2},$$

giving the initial representative point

$$x^{(0)}_e := a_e + \ell^*_e\, w_e(\xi^*_e). \tag{4.3}$$

Because ℓ^*_e lies strictly between the open-interval endpoints, $x^{(0)}_e \in \Omega^\circ$.

Containment in Ω° does not guarantee a valid voxel index: gate certification, interval evaluation, midpoint construction, and voxel-index validation are separate finite-precision computations, and success at one stage does not imply success at the next. An event whose midpoint procedure fails to produce a strict-interior point, or whose constructed point is rejected by voxel-index validation, is dropped. The set delivered to the reconstruction engine is

$$\widehat{E}^{\text{init}}_{K_{\text{az}}} := \left\{ e \in \widehat{E}^{\text{gate}}_{K_{\text{az}}} : x^{(0)}_e \text{ is constructed and accepted} \right\},$$

with cardinality $N_{\text{init}} := |\widehat{E}^{\text{init}}_{K_{\text{az}}}|$. In the exact geometric formulation, a sampled strict-interior witness is a sufficient certificate of usability. Thus the certified implementation set is a conservative subset of the theoretical surviving-event set,

$$\widehat{E}^{\text{init}}_{K_{\text{az}}} \subseteq \widehat{E}^{\text{gate}}_{K_{\text{az}}} \subseteq E.$$

The implementation therefore does not redefine event-wise admissibility. It changes only the finite event pool used by the run: events may be lost because usability was not certified by the sampled gate, or because the constructed initial point failed finite-precision validation.

Reconstruction is then performed on this certified subset rather than on the full theoretical set E. The realised state space is therefore

$$\widehat{\mathcal{S}} := \prod_{e \in \widehat{E}^{\text{init}}_{K_{\text{az}}}} \Sigma_e,$$

so the chain is realised on $\widehat{\mathcal{S}}$.

Every event in $\widehat{E}^{\text{init}}_{K_{\text{az}}}$ carries a constructed representative point $x_e^{(0)}$ that lies strictly inside the VOI, is accepted by voxel-index validation, and satisfies the admissibility criterion for event e. These representatives therefore define the implementation's initial chain state

$$\mathbf{s}^{(0)} := \left(x_e^{(0)}\right)_{e \in \widehat{E}^{\text{init}}_{K_{\text{az}}}},$$

which is admissible in $\widehat{\mathcal{S}}$. If $N_{\text{init}} = 0$, no admissible initial state exists and reconstruction terminates before the chain begins.

4.3 Reconstruction engine

The reconstruction engine operates on the certified event set $\widehat{E}^{\text{init}}_{K_{\text{az}}}$, the admissible initial state $\mathbf{s}^{(0)}$, and the validated run configuration. Its task is to realise the single-event Markov chain on the finite runtime state space $\widehat{\mathcal{S}}$.

The engine preserves the SOE update structure: one event representative is selected, a new admissible origin is proposed for that event, and the move is accepted or rejected using the current occupancy image [18, 20]. What changes at implementation level is not the mathematical support Σ_e, but the finite realisation: event indices, proposal primitives, voxel indices, occupancy updates, and state mutations are represented explicitly in memory.

4.3.1 Proposal generation

The proposal routine implements the within-event proposal law defined in section 3.4. In the SOE literature, the update step selects an event and proposes a new admissible origin for that event; for cone-based SOE this means proposing a new point on the event's conical surface inside the reconstruction region [20, 21]. Andreyev et al. emphasise that random locations on the half-cone surface must be uniformly distributed, since nonuniform location generation may bias the reconstructed image. They also note that the rejection-based strategy is inefficient when only a small part of the cone intersects the VOI, and leave more efficient generation on the VOI-intersecting cone portion for future work [20].

The direct clipped-cone sampler used here addresses that proposal-generation problem. In the present formulation, the uniform-location requirement is realised as a surface-area-uniform proposal law on the clipped feasible surface Σ_e. The bounded-box cone-surface decomposition, segment masses, The inverse-CDF azimuth draw, and radial square-root law are therefore implementation choices of this work, with the derivation given in appendix A.5.

Each step selects an internal runtime index

$$r \in \{0, \dots, N_{\text{init}} - 1\}$$

uniformly at random from a dedicated seeded stream and proposes a new representative point on the feasible surface of the corresponding event. Let $r \mapsto e_r$ denote the fixed runtime enumeration of $\widehat{E}^{\mathrm{init}}_{K_{\mathrm{az}}}$.

For the selected event e_r, the proposal routine builds or reuses cached sampling primitives. These are the visible azimuth segments, i.e. azimuth intervals with nonempty clipped radial support, together with their surface-area masses under the cone parametrisation. The inverse-CDF construction that converts a uniform variate into a segment-conditional azimuth is evaluated internally at draw time.

The candidate draw has two stages. In the first, the sampler selects a visible azimuth segment with probability proportional to its surface-area mass and draws an azimuth ξ on that segment by inverting the segment-conditional CDF. The sampled azimuth fixes the generator direction $w_{e_r}(\xi)$.

In the second stage, conditional on ξ, the sampler draws a radial coordinate from the surface-area conditional on that generator. The proposed representative point is

$$x'_r = a_{e_r} + \ell\, w_{e_r}(\xi),$$

where

$$\ell = \sqrt{\left(\ell^-_{e_r}(\xi)\right)^2 + U\left[\left(\ell^+_{e_r}(\xi)\right)^2 - \left(\ell^-_{e_r}(\xi)\right)^2\right]}, \qquad U \sim \mathrm{Unif}(0,1),$$

and $\ell^-_{e_r}(\xi)$, $\ell^+_{e_r}(\xi)$ are the radial support limits at azimuth ξ. The square-root form follows from the cone-surface area element; the full derivation is given in appendix A.5. This two-stage procedure produces draws that are uniform in surface area on Σ_{e_r}.

Before the proposal is returned, the implementation validates the sampled point with the reconstruction admissibility predicate. This check is a finite-precision safeguard: in exact arithmetic, the clipped-cone sampler already returns points in Σ_{e_r}. Only a soft numerical cone-surface miss triggers redraws, up to the configured retry budget ($R_{\max} := 32$). Hard failures, such as failed VOI containment or voxel-index validation, raise an internal sampler error.

This validation is not a modelling rejection step. Except for soft finite-precision cone-surface misses, a failed check indicates inconsistency between the sampling primitives and the admissibility predicate rather than a draw from a modified proposal law. The returned proposal therefore carries only the candidate point.

4.3.2 Acceptance and state update

The acceptance routine implements the local ratio derived in section 3.5. Because a single proposal changes only one event representative, the update can be resolved from the departure voxel, the arrival voxel, and the two affected occupancies.

The kernel receives the certified candidate point x'_r and resolves the acceptance decision through local lookups. The departure voxel v^- is read from the cached voxel assignment in the mutable runtime state. The arrival voxel v^+ is computed by voxelising the candidate point; the equality test $v^+ = v^-$ is evaluated only after both indices are available.

If $v^+ = v^-$, the move is accepted unconditionally. If $v^+ \neq v^-$, the kernel reads the current occupancies $n_{v^-}(\mathbf{s})$ and $n_{v^+}(\mathbf{s})$ from the occupancy array and evaluates the acceptance ratio (3.11). The implementation fixes $\alpha = 1$; this value is not exposed as a runtime parameter. A uniform draw from the dedicated seeded stream decides acceptance.

On acceptance with $v^+ \neq v^-$, three mutations are applied as one accepted update: the occupancy array is decremented at v^- and incremented at v^+, and both the representative point and the cached voxel index for runtime index r are overwritten. On acceptance with $v^+ = v^-$, only the representative point is overwritten. On rejection, no mutation occurs. After every step, the occupancy array matches the voxel histogram of the current cached assignments.

The chain loop repeats this cycle until the configured number of steps is exhausted.

4.4 Output, verification, and runtime measurement

A completed reconstruction run produces a fixed set of persisted artifacts and enforces implementation invariants through fail-fast checks. This section also states the runtime measurement protocol used for the reconstruction timings reported later.

4.4.1 Persisted artifacts

The implementation writes six artifacts to a single output directory. Two are array files, one is the formal output contract, and three are diagnostic sidecars.

The primary array is the retained-state mean occupancy $\bar{n}_v$ (3.12), accumulated online during chain execution and persisted as a float array of shape (n_x, n_y, n_z). The accumulator adds the current occupancy field to a running sum at each retained step and divides by C at termination, so no post-pass recomputation over stored states is required.

The secondary array is the terminal-state occupancy field $\mathbf{n}(\mathbf{s}^{(K)})$ (3.13), persisted as an integer array of the same shape. Together, the two arrays provide the averaged reconstruction and a single-draw snapshot of the chain endpoint.

The output contract is a metadata record that binds the two arrays to their reconstruction geometry: VOI bounds, grid shape, voxel spacings, and axis order. It also records chain provenance, burn-in length B, thinning interval T, retained count C, total steps K, and the terminal step index, so that any downstream consumer can interpret the arrays without access to the run configuration.

The first diagnostic sidecar records run instrumentation: event counts at each preprocessing stage, chain step and acceptance counts, wall-clock timings, and cheap summary statistics of the emitted arrays.

The second records online Metropolis–Hastings health summaries: proposal and acceptance rates, occupancy-count histograms, and departure–arrival count-pair statistics.

The third records the certified event pool used to seed the chain, using stable runtime/admitted-event identities and the admission settings under which certification was performed. Raw upstream identifiers from the input file are not retained in this artifact.

4.4.2 Runtime invariant checks

The implementation enforces its invariants through entry checks and pointwise runtime assertions. The strategy is fail-fast: a violated precondition raises an error and aborts the run. These checks do not prove convergence or reconstruction accuracy; they ensure that the finite runtime state remains consistent with the voxelised state representation assumed by the kernel.

Before the chain begins, the entry check confirms that the supplied occupancy array has the correct shape and integer type, is elementwise non-negative, and exactly matches the voxel histogram induced by the initial representative points. If any of these conditions fails, the run aborts before the first step.

During chain execution, the occupancy array is maintained by local decrement–increment updates. The local-update rule in section 4.3.2 is therefore checked against the occupancy invariant during execution. If a proposal produces a point that is not voxelisable, or if a decrement would reduce a voxel count below zero, the implementation raises an error.

4.4.3 Runtime measurement protocol

The timing protocol records the computational cost of the reconstruction engine under the same configuration used for the reported range-verification runs. All timed runs used single-process Python on an Apple M3 system with 8 GB of unified memory. The reconstruction grid contained $10 \times 15 \times 150$ voxels, giving 22,500 voxels in total.

Sampled event admission used $K_{\mathrm{az}} = 512$ azimuthal samples. Each chain ran for 1000 sweeps, with burn-in $B = 250$ sweeps and thinning interval equal to one sweep. The retained ensemble therefore contained $C = 750$ post-burn-in states. These settings were fixed across the timed energy configurations, so differences in measured runtime reflect the admitted event count and the corresponding number of single-event update attempts.

4.5 Offline analysis pipeline

The offline analysis pipeline fixes how persisted reconstruction bundles and PHITS truth files are reduced to the quantities reported in chapter 5. It defines the analysed beam-energy cohort, truth inputs, reconstruction domain, feature-extraction conventions, calibration convention, reconstructed error summaries, and geometry-only diagnostics.

4.5.1 Analysed inputs, reconstruction domain, and cohorts

The analysed data set consists of eight beam energies:

$$100,\ 105,\ 110,\ 115,\ 120,\ 125,\ 135,\ 150\ \text{MeV}.$$

All event records and truth profiles in this cohort are PHITS outputs, with the neutron nuclear-data setting documented as Japanese Evaluated Nuclear Data Library, version 4.0 (JENDL-4.0) [36]. For each energy, table 4.1 lists the PHITS truth inputs and their roles in the analysis.

Table 4.1: Truth inputs and their roles in the analysis.

Truth file	Use	x aperture / bins	y aperture / bins	z bounds / bins
`product_z_n-events.pickle.xz`	Operational reference: distal R_{50} of the neutron-event production profile	$[-2, 2]$ cm; 1 bin	$[-2, 2]$ cm; 1 bin	$[-20, 20]$ cm; 201 bins
`product_z_all-with-scatters.pickle.xz`	Fast-neutron product-z truth; neutron row selected	$[-15, 15]$ cm; 1 bin	$[-15, 15]$ cm; 1 bin	$[-20, 20]$ cm; 201 bins
`xyz_product_n-events-with-scatters.pickle.xz`	Three-dimensional fast-neutron truth for spatial projections	$[-15, 15]$ cm; 151 bins	$[-15, 15]$ cm; 151 bins	$[-15, 1]$ cm; 81 bins

The two one-dimensional truth files serve different roles. The no-scatter neutron-event production profile supplies the operational reference depth, whereas the with-scatter fast-neutron product-depth profile supplies the surrogate features evaluated against that reference. The PHITS runs were configured for up to 100 000 000 primary source histories, but not all energies completed the full batch count. Table 4.2 therefore reports the completed source histories and the raw neutron cone-event cohorts available before filtering and preprocessing. The post-filtering event count used by each reconstruction run is reported in table 5.2.

Table 4.2: Completed source histories and neutron cone-event cohorts.

Beam energy	Completed source protons	Cone count
100 MeV	100 000 000	4248
105 MeV	81 000 000	3867
110 MeV	78 000 000	4066
115 MeV	75 000 000	4270
120 MeV	70 000 000	4314
125 MeV	68 000 000	4668
135 MeV	61 000 000	4840
150 MeV	75 000 000	7114

All truth comparisons use centre-coordinate grids implied by the metadata arrays.[1] The product-depth profiles are analysed on their native transverse apertures; no transverse matching is applied before depth-feature extraction. Consequently, depth features are compared under the stated calibration convention rather than after transverse aperture equalisation.

[1] PHITS-Tools stores edge arrays with the upper edge one bin beyond the declared maximum.

Reconstruction domain and grids

All reconstruction runs share a common VOI:

$$x \in [-15, 15] \text{ cm}, \qquad y \in [-20, 20] \text{ cm}, \qquad z \in [-15, 15] \text{ cm}.$$

The beam-axis unit vector is $b = (0, 0, 1)$, so distal depth increases with z. Table 4.3 lists the three grids used.

Table 4.3: Reconstruction grids used in the offline analysis.

Run grid	Bins (x, y, z)	Voxel size (x, y, z) (cm)	Total voxels
$10 \times 15 \times 150$	$(10, 15, 150)$	$(3.0,\ 2.6667,\ 0.2)$	22 500
$18 \times 24 \times 150$	$(18, 24, 150)$	$(1.6667,\ 1.6667,\ 0.2)$	64 800
$20 \times 30 \times 150$	$(20, 30, 150)$	$(1.5,\ 1.3333,\ 0.2)$	90 000

Several candidate reconstruction grids were tested during method development. The $10 \times 15 \times 150$ grid gave the most stable overall behaviour among the tested coarser and finer candidates, and it is therefore used as the native grid for the main analysis. The two finer retained grids are used only as grid-comparison controls. The analysed runs are grouped by purpose: the main range-verification cohort, the grid-comparison cohort, the sweep-depth sensitivity cohort, and an event-multiplicity control for sparsity stress testing.

Analysed cohorts

The main range-verification cohort contains native reconstructions on the $10 \times 15 \times 150$ grid at 1000 sweeps. Three seed configurations, S0–S2, fix the initialisation, proposal, and acceptance random streams at each beam energy, giving 24 reconstructions. The same seed labels are reused across energies so that seed-wise variation is not confounded with beam energy.

The grid-comparison cohort uses the $10 \times 15 \times 150$, $18 \times 24 \times 150$, and $20 \times 30 \times 150$ grids at 1000 sweeps for 100 MeV, 115 MeV, 135 MeV, and 150 MeV. These runs support the within-energy occupancy response in fig. 5.5, the three-grid mean reported for reconstructed peak-depth tracking in fig. 5.7, and the reconstructed peak-depth entries in table 5.6.

Sweep-depth sensitivity is evaluated at 100 MeV on the $10 \times 15 \times 150$ grid using 500, 1000, 2000, 5000, and 10 000 sweeps with seed S0. Each sweep count defines one run, so no seed averaging is applied.

An event-multiplicity control pairs the native 100 MeV, $10 \times 15 \times 150$, 2000-sweep S0 run with a matched run constructed by repeating the same cone-event pool to 50 000 event entries. The control changes effective event multiplicity without creating an independent physical dataset, so its role is limited to a sparsity stress test.

4.5.2 Truth-profile extraction and calibration protocol

The calibration protocol is truth-to-truth: both input profiles are PHITS-derived neutron truth profiles. Its purpose is to map candidate fast-neutron depth features onto a stable operational falloff-depth reference before the same feature rules are applied to reconstructed profiles.

The reference profile is the no-scatter neutron-event production profile. In this water-phantom dataset, the distal support of this production field terminates at the simulated primary-beam stopping depth, so the endpoint was also tested as a reference target. Endpoint calibration is the more direct physical range choice, but the best truth-to-truth endpoint calibrator was the with-scatter neutron R_{20}. This feature was not consistently defined in the reconstructed profiles because their distal tails did not fall below 20% within the reconstructed volume. The distal R_{50} of the no-scatter profile was therefore retained as the operational falloff-depth reference. It provides a stable distal marker for comparing range-related shifts across beam energies while remaining compatible with features extractable from the reconstructed profiles.

The surrogate profile is the neutron component of the with-scatter product-depth tally, which is closer to the field supplied to reconstruction. The calibration maps depth features from this surrogate profile onto the operational falloff-depth reference. Both profiles are reduced to one-dimensional axial profiles by marginalising over the transverse tally dimensions within each z bin.

Normalisation and validity convention

A profile is valid when all entries are finite and its maximum is positive. Valid profiles are normalised by dividing by their maximum. All analysed truth profiles pass this validity check; reconstructed profiles that fail extraction are marked invalid.

Truth-feature extraction

The feature extraction rule defines which axial markers are tested as possible surrogates for the operational falloff-depth reference. The candidate fast-neutron surrogate features are extracted from the normalised fast-neutron with-scatter product-depth profile:

$$z_{\mathrm{peak}}^{\mathrm{truth}}(E_b), \qquad R_{50}^{\mathrm{truth}}(E_b), \qquad R_{20}^{\mathrm{truth}}(E_b), \qquad R_{90}^{\mathrm{truth}}(E_b).$$

Peak depth is the mean axial centre coordinate of all bins attaining the normalised maximum. For $q \in \{20, 50, 90\}$, R_q is the first distal downward crossing of the normalised profile through level $q/100$ after the profile maximum, computed by linear interpolation between adjacent axial bin centres. If no such crossing is bracketed, the feature is invalid.

Truth-to-truth calibration

For each candidate feature f, let $d_{\mathrm{truth}}^{(f)}(E_b)$ denote the depth returned by applying the feature extraction rule for f to the with-scatter product-depth profile at beam energy E_b. The valid-energy set is

$$\mathcal{E}_f := \{E_b : d_{\mathrm{truth}}^{(f)}(E_b) \text{ is defined}\}.$$

Absolute reference calibration fits a linear map from the with-scatter feature depth to the operational falloff-depth reference:

$$\hat{p}^{(f)}(E_b) = A_f + B_f\, d_{\mathrm{truth}}^{(f)}(E_b),$$

where A_f and B_f are determined by ordinary least squares over $\mathcal{E}_f$ against $p_{\text{ref}}(E_b)$. The resulting residual is

$$e^{(f)}(E_b) = \hat{p}^{(f)}(E_b) - p_{\text{ref}}(E_b), \qquad E_b \in \mathcal{E}_f.$$

Because the same energy set is used to fit and summarise the calibration, these are in-sample calibration residuals.

Calibration quality is summarised over $\mathcal{E}_f$ by

$$\text{RMSE}^{(f)} = \sqrt{\frac{1}{|\mathcal{E}_f|}\sum_{E_b\in\mathcal{E}_f} e^{(f)}(E_b)^2}, \qquad \text{MaxAE}^{(f)} = \max_{E_b\in\mathcal{E}_f} |e^{(f)}(E_b)|.$$

Pearson correlation between $d^{(f)}_{\text{truth}}(E_b)$ and $p_{\text{ref}}(E_b)$ is computed over $\mathcal{E}_f$.

Candidate features are screened before ranking. A feature is truth-admissible only if it is defined at every analysed energy and varies monotonically with p_{ref}; ties do not violate monotonicity. Truth-admissible features are ranked by increasing RMSE, with MaxAE as the tie-break. The highest-ranked truth-admissible feature defines the primary truth-level surrogate for the operational falloff-depth reference.

Truth-level admissibility does not by itself make a feature usable for calibrated reconstruction outputs. A truth-admissible feature is propagated only if the same feature has at least one valid reconstructed seed at every analysed energy in the main range-verification cohort. Energy-wise summaries then use the valid seed count for that feature and energy. Features that pass truth-level screening but fail this propagation criterion are retained as profile-shape diagnostics, but are not used to report calibrated depth estimates.

This separates truth-level feature selection from the stricter requirement that the same feature can be extracted from reconstructed profiles across the main analysis cohort.

Truth-level shifts and compression

To characterise how each surrogate feature shifts with beam energy before calibration, shifts are reported relative to the lowest analysed energy:

$$\Delta d^{(f)}_{\text{truth}}(E_b) = d^{(f)}_{\text{truth}}(E_b) - d^{(f)}_{\text{truth}}(100\ \text{MeV}).$$

The cross-energy span

$$D_f = \max_{E_b} d^{(f)}_{\text{truth}}(E_b) - \min_{E_b} d^{(f)}_{\text{truth}}(E_b)$$

quantifies the total excursion of feature f. The corresponding reference span is

$$D_{p_{\text{ref}}} = \max_{E_b} p_{\text{ref}}(E_b) - \min_{E_b} p_{\text{ref}}(E_b).$$

Since the operational falloff-depth reference varies across the analysed energies, $D_{p_{\text{ref}}} > 0$. For each feature f, the truth-level compression ratio is

$$C^{(f)}_{\text{truth}} = \frac{D_f}{D_{p_{\text{ref}}}}.$$

This ratio measures how much of the reference-depth dynamic range the surrogate resolves: values near unity indicate one-to-one depth tracking, whereas values well below unity indicate compression into a smaller interval.

4.5.3 Reconstruction-output reductions

The reconstruction-output reductions apply the truth-level feature rules from section 4.5.2 to retained reconstruction fields. They convert each reconstructed volume into axial profiles, calibrated depth estimates, error summaries, dynamic-range diagnostics, and spatial projections.

Axial profile and reconstructed features

For a fixed beam energy E_b and reconstruction seed s, let $\bar{n}_{ijk,s}(E_b)$ denote the retained mean field after axis-order restoration, with $i \in I_x$, $j \in I_y$, and $k \in I_z$. The reconstructed axial profile is the transverse marginal

$$h_{k,s}(E_b) := \sum_{i\in I_x}\sum_{j\in I_y} \bar{n}_{ijk,s}(E_b), \qquad k \in I_z.$$

This marginalisation produces the one-dimensional reconstructed profile to which the truth-profile validity, normalisation, and feature-extraction rules can be applied. The reconstruction pipeline extracts the propagated features from each valid $h_{k,s}(E_b)$, subject to the propagation rule in section 4.5.2.

Range conversion and error summaries

The following summaries are defined for propagated features in the main range-verification cohort. For $f \in \{\text{peak}, R_{50}\}$, let $d^{(f)}_{\text{rec},s}(E_b)$ denote the reconstructed depth of feature f from seed s at beam energy E_b. At each energy, $N_s^{(f)}(E_b)$ denotes the number of valid seed reconstructions for feature f. For propagated features, this number is nonzero at every analysed energy, but it may vary with feature and energy.

The truth-calibration coefficients map each reconstructed feature depth to a reconstructed estimate of the operational falloff-depth reference:

$$\hat{p}^{(f)}_{\text{rec},s}(E_b) = A_f + B_f\, d^{(f)}_{\text{rec},s}(E_b).$$

The corresponding seed-level reference-depth error is

$$e^{(f)}_{\text{rec},s}(E_b) = \hat{p}^{(f)}_{\text{rec},s}(E_b) - p_{\text{ref}}(E_b).$$

Positive error denotes overestimation relative to the operational falloff-depth reference.

At each energy, the valid seed predictions define the seed-mean reconstructed reference-depth estimate

$$\bar{p}^{(f)}_{\text{rec}}(E_b) = \frac{1}{N_s^{(f)}(E_b)} \sum_s \hat{p}^{(f)}_{\text{rec},s}(E_b),$$

where the sum is over valid seeds. The corresponding signed seed-mean error is

$$\bar{e}^{(f)}(E_b) = \bar{p}^{(f)}_{\text{rec}}(E_b) - p_{\text{ref}}(E_b).$$

Seed spread is the sample standard deviation of the valid seed predictions and is reported when $N_s^{(f)}(E_b) \geq 2$. Energy-resolved tables report $\bar{p}^{(f)}_{\text{rec}} \pm \sigma^{(f)}_{\text{seed}}$ together with $\bar{e}^{(f)}$.

The preceding quantities are energy-resolved. Aggregate summaries collapse these energy-level errors across the analysed energies for each propagated feature.

Aggregate reconstruction errors are computed over

$$\mathcal{A}_f = \{E_b : \bar{p}^{(f)}_{\text{rec}}(E_b) \text{ is valid}\}.$$

The reconstruction RMSE is the RMSE of the signed seed-mean errors $\bar{e}^{(f)}$ over $\mathcal{A}_f$. This measures the error of the seed-averaged reconstructed estimate. The absolute-error summaries use the per-energy mean absolute seed error

$$m^{(f)}_{\text{abs}}(E_b) = \frac{1}{N^{(f)}_s(E_b)} \sum_s |e^{(f)}_{\text{rec},s}(E_b)|,$$

where the sum is over valid seeds. Thus

$$\text{MAE}^{(f)}_{\text{rec}} = \frac{1}{|\mathcal{A}_f|} \sum_{E_b \in \mathcal{A}_f} m^{(f)}_{\text{abs}}(E_b), \qquad \text{MaxMeanAE}^{(f)}_{\text{rec}} = \max_{E_b \in \mathcal{A}_f} m^{(f)}_{\text{abs}}(E_b).$$

Single-seed sweep-depth summaries report $|e^{(f)}_{\text{rec},s}(E_b)|$. Aggregate seed-spread summaries are the mean and maximum of $\sigma^{(f)}_{\text{seed}}(E_b)$ over $E_b \in \mathcal{A}_f$.

Dynamic range and monotonicity

For each propagated feature, the reconstructed reference-depth span is computed from the seed-mean calibrated estimates:

$$D^{(f)}_{\text{pred}} = \max_{E_b \in \mathcal{A}_f} \bar{p}^{(f)}_{\text{rec}}(E_b) - \min_{E_b \in \mathcal{A}_f} \bar{p}^{(f)}_{\text{rec}}(E_b).$$

The reconstructed span ratio is

$$C^{(f)}_{\text{span}} = \frac{D^{(f)}_{\text{pred}}}{D_{p_{\text{ref}}}}.$$

The reconstructed calibrated response is monotone when the seed-mean estimates preserve the beam-energy ordering over $\mathcal{A}_f$; ties are allowed.

Spatial projections

For each selected energy–grid case, runs with matching array shape are averaged elementwise. The coordinate metadata from the first run in the group define the reconstruction centre-coordinate grid.

The seed-averaged reconstruction volume is compared with the with-scatter fast-neutron truth volume on a common centre-coordinate support. This support is constructed per axis by intersecting the reconstruction and truth centre-coordinate ranges.

For a cropped reconstruction or truth subvolume g_{ijk}, the displayed projections are

$$P^{xz}_{ik} := \sum_j g_{ijk}, \qquad P^{yz}_{jk} := \sum_i g_{ijk}.$$

Each projected image is normalised by its valid positive maximum before plotting. Projections without such a maximum are masked.

4.5.4 Geometry-only axial accessibility analysis

The geometry-only analysis isolates the axial information contained in the certified cone geometry before SOE dynamics are applied. It has three levels. First, each admitted cone is reduced to an eventwise axial area law. Second, the sampled axial footprint of each cone is summarised by width, uniformity, and axis-alignment diagnostics. Third, the eventwise laws are averaged into an aggregate geometry-only axial profile whose cross-energy span can be compared with the operational reference span.

Write

$$\widehat{E}_{\mathrm{gate}} := \widehat{E}^{\mathrm{gate}}_{K_{\mathrm{az}}^{\mathrm{gate}}}$$

for the certified admission set from section 4.2.2. The diagnostic uses the same sampled-generator-direction and ray–VOI-intersection construction as the admission gate, but with a separate area quadrature. Instead of applying only the binary admission predicate, it accumulates clipped cone-surface area into axial bins. The area quadrature uses $K_{\mathrm{az}}^{\mathrm{area}} = 1024$.[2]

For $e \in \widehat{E}_{\mathrm{gate}}$ and $\xi \in \Xi^{(K_{\mathrm{az}}^{\mathrm{area}})}$, the closure of the open radial support is written

$$[\ell_e^-(\xi),\, \ell_e^+(\xi)] := \overline{I_e^\circ(\xi)}$$

whenever $I_e^\circ(\xi) \neq \varnothing$. The strict-open support keeps the diagnostic tied to the admission convention; its closure then provides the radial interval used for area accumulation. The retained diagnostic azimuths are

$$\Xi_e^{\mathrm{ret}} := \left\{ \xi \in \Xi^{(K_{\mathrm{az}}^{\mathrm{area}})} : I_e^\circ(\xi) \neq \varnothing \right\}.$$

The diagnostic-valid event set is

$$\widehat{E}_{\mathrm{area}} := \left\{ e \in \widehat{E}_{\mathrm{gate}} : \Xi_e^{\mathrm{ret}} \neq \varnothing \right\}, \qquad N_{\mathrm{area}} := |\widehat{E}_{\mathrm{area}}|.$$

Admitted events outside $\widehat{E}_{\mathrm{area}}$ are invalid for the area diagnostic and do not contribute to the geometry-only summaries.

For one retained azimuth, the clipped cone-strip area is

$$\Delta A_e(\xi) = \frac{\pi \sin\theta_e}{K_{\mathrm{az}}^{\mathrm{area}}} \left(\ell_e^+(\xi)^2 - \ell_e^-(\xi)^2 \right).$$

This is the conical surface-area element integrated over the sampled azimuth width and the clipped radial interval.

The strip areas are then distributed over axial bins to form an eventwise depth law.

Let Z_k denote the axial slab of bin k, using the same half-open bin convention as the reconstruction grid. For a retained azimuth and axial bin, let $[\alpha_{e,k}(\xi), \beta_{e,k}(\xi)]$ be the part of $[\ell_e^-(\xi), \ell_e^+(\xi)]$ for which $a_{e,z} + \ell w_{e,z}(\xi) \in Z_k$. If this intersection is empty, the bin contribution is zero. Otherwise,

$$\Delta A_{e,k}(\xi) = \frac{\pi \sin\theta_e}{K_{\mathrm{az}}^{\mathrm{area}}} \left(\beta_{e,k}(\xi)^2 - \alpha_{e,k}(\xi)^2 \right).$$

[2] Repeating the diagnostic with $K_{\mathrm{az}}^{\mathrm{area}} = 512$ did not change the reported geometry-level conclusions.

For $w_{e,z}(\xi) = 0$, the same half-open bin convention assigns the whole strip to the axial bin containing $a_{e,z}$. Summing the partial strip areas gives the eventwise axial area histogram

$$A_{e,k} := \sum_{\xi \in \Xi_e^{\mathrm{ret}}} \Delta A_{e,k}(\xi), \qquad k \in I_z.$$

The eventwise axial accessibility law is the unit-sum normalisation

$$g_{e,k} := \frac{A_{e,k}}{\sum_{\ell \in I_z} A_{e,\ell}}, \qquad k \in I_z.$$

Thus $g_{e,\cdot}$ is the axial probability law induced by one admitted cone inside the VOI.

The same retained intervals also support event-level diagnostics that describe how broadly and how uniformly each cone covers the axial direction. Its lower and upper limits are

$$z_e^- := \min_{\xi \in \Xi_e^{\mathrm{ret}}} \min\bigl\{a_{e,z} + \ell_e^-(\xi)\, w_{e,z}(\xi),\ a_{e,z} + \ell_e^+(\xi)\, w_{e,z}(\xi)\bigr\},$$

$$z_e^+ := \max_{\xi \in \Xi_e^{\mathrm{ret}}} \max\bigl\{a_{e,z} + \ell_e^-(\xi)\, w_{e,z}(\xi),\ a_{e,z} + \ell_e^+(\xi)\, w_{e,z}(\xi)\bigr\}.$$

Both expressions evaluate the z-coordinate at each radial endpoint. The minimum and maximum over retained azimuths bound the sampled axial footprint. The axial support width is

$$W_e := z_e^+ - z_e^-,$$

and the reported width fraction is W_e/L_z, where L_z is the axial extent of the VOI.

The uniform footprint reference is defined only for events with $W_e > 0$. Events with $W_e = 0$ are excluded from the footprint-uniformity summaries. For the remaining events, the uniform reference on the sampled axial footprint is the unit-normalised overlap length

$$m_{e,k}^{\mathrm{unif}} := \frac{|Z_k \cap [z_e^-, z_e^+]|}{W_e}, \qquad k \in I_z.$$

The eventwise total variation distance from this reference is

$$\mathrm{TV}_e = \frac{1}{2} \sum_{k \in I_z} \bigl| g_{e,k} - m_{e,k}^{\mathrm{unif}} \bigr|.$$

Small TV_e means that the event distributes cone-surface area nearly uniformly over its sampled axial footprint.

The reported event-fraction summaries use the descriptive thresholds

$$\frac{W_e}{L_z} > 0.50, \qquad \mathrm{TV}_e < 0.10.$$

The event-axis alignment statistic is

$$|u_e \cdot b|.$$

Values near zero correspond to transverse cone axes, while values near one correspond to beam-aligned cone axes. This statistic is reported as an event-level geometry diagnostic.

Per-energy eventwise summaries report medians over $\widehat{E}_{\text{area}}$. Percentage summaries use N_{area} as the denominator, except for footprint-uniformity summaries, which use only events with $W_e > 0$.

After the eventwise diagnostics, the geometry-only axial response is summarised by averaging the eventwise accessibility laws.

The aggregate geometry-only axial profile is the equal-event-weighted mean

$$G_k := \frac{1}{N_{\text{area}}} \sum_{e \in \widehat{E}_{\text{area}}} g_{e,k}, \qquad k \in I_z.$$

Because each $g_{e,\cdot}$ has unit mass, G_k gives each diagnostic-valid admitted event equal weight rather than weighting events by total clipped cone-surface area. The geometry-only peak depth $z_{\text{peak}}^{\text{geom}}$ is extracted from G_k by the peak rule in section 4.5.2, evaluated on the axial centre grid. Its cross-energy span is

$$D_{z_{\text{peak}}^{\text{geom}}} := \max_{E_b} z_{\text{peak}}^{\text{geom}}(E_b) - \min_{E_b} z_{\text{peak}}^{\text{geom}}(E_b).$$

The geometry-only compression ratio is

$$C_{\text{geom}} := \frac{D_{z_{\text{peak}}^{\text{geom}}}}{D_{p_{\text{ref}}}},$$

where $D_{p_{\text{ref}}}$ is the span of the operational falloff-depth reference. This ratio measures how much of the operational reference-depth dynamic range the cone geometry alone preserves before SOE dynamics are applied.

4.5.5 Chain-health reduction and transition displays

The chain-health reduction uses the persisted `chain_health.json` sidecar to summarise cross-voxel proposal dynamics after reconstruction. Its purpose is diagnostic: it records what occupancy transitions the chain attempted and accepted, not whether the chain has converged. Same-voxel proposals are excluded from this reduction because their acceptance probability is one and they do not test the occupancy-imbalance rule. The departure–arrival table records attempted moves from a departure voxel with pre-move occupancy $i \geq 1$ to an arrival voxel with pre-move occupancy $j \geq 0$. Here i and j denote $n_{v^-}(\mathbf{s})$ and $n_{v^+}(\mathbf{s})$, respectively. The recorded proposal and accepted counts are assembled on the observed count range as

$$N_{ij}^{\text{prop}}, \qquad N_{ij}^{\text{acc}}.$$

Row $i = 0$ is unused because a selected departure voxel must contain at least one representative point.

For cross-voxel proposals at $\alpha = 1$, the acceptance rule from section 3.5 gives probability one when $i \leq j + 1$ and $(j + 1)/i$ otherwise. The empirical acceptance heatmap is interpreted against this occupancy-imbalance threshold.

The normalisation scale for the count-pair table is the total attempted mass represented in the sidecar,

$$T_{\text{att}} := \sum_{i \geq 1} \sum_{j \geq 0} N_{ij}^{\text{prop}}.$$

This scale defines the empirical proposal mass and conditional acceptance rate,

$$\widehat{p}_{ij} := \frac{N_{ij}^{\mathrm{prop}}}{T_{\mathrm{att}}}, \qquad \widehat{a}_{ij} := \frac{N_{ij}^{\mathrm{acc}}}{N_{ij}^{\mathrm{prop}}} \quad \text{for } N_{ij}^{\mathrm{prop}} > 0.$$

The proposal mass $\widehat{p}_{ij}$ sums to one over the recorded count-pair grid. The rate $\widehat{a}_{ij}$ is defined only for count pairs with at least one proposal.

Low-count summaries are computed from the proposal mass. The reported marginal and cell masses are

$$\mathbb{P}_{\widehat{p}}(i=1) := \sum_{j\geq 0} \widehat{p}_{1j}, \qquad \mathbb{P}_{\widehat{p}}(j=0) := \sum_{i\geq 1} \widehat{p}_{i0}, \qquad \mathbb{P}_{\widehat{p}}((1,0)) := \widehat{p}_{10},$$

and the low-arrival imbalance mass is

$$\mathbb{P}_{\widehat{p}}(j < i-1) := \sum_{i\geq 2} \sum_{j=0}^{i-2} \widehat{p}_{ij}.$$

Transition heatmaps use fixed display masks. Proposal and accepted-flow masses are plotted on logarithmic scales, so nonpositive cells are masked. Acceptance rates are masked when $N_{ij}^{\mathrm{prop}} < 5$.

Single-run transition diagnostics use one anchor reconstruction. The anchor is the native 100 MeV, $10 \times 15 \times 150$, 1000-sweep reconstruction with seed configuration S0.

4.5.6 Cohort aggregation and comparison rules

Cohort-level summaries use the analysed cohorts defined in section 4.5.1 and the reconstructed observables defined in section 4.5.3. Each aggregation rule is tied to one comparison question. Run-level diagnostic summaries therefore retain one row per reconstruction, and no averaging is applied across runs unless an averaging rule is stated explicitly.

For single-seed sweep-depth runs, $\bar{e}^{(f)}(E_b) = e^{(f)}_{\mathrm{rec},S0}(E_b)$, and the reported absolute calibrated reference-depth error is $|e^{(f)}_{\mathrm{rec},S0}(E_b)|$.

Control plots use a deduplicated reconstruction table when a comparison requires one representative per energy–grid pair. The representative is selected at 1000 sweeps S0. Event-multiplicity paired comparisons do not use this deduplicated table; they retain matched rows at fixed energy, grid, sweep count, and seed.

Cross-grid peak tracking uses the deduplicated representatives from the $10 \times 15 \times 150$, $18 \times 24 \times 150$, and $20 \times 30 \times 150$ grids. At each included energy, the reported uncalibrated reconstructed peak depth is the arithmetic mean of the three deduplicated peak depths. This averaging rule applies only to cross-grid peak tracking, not to seed-resolved range-verification metrics.

Within-energy sparsity-control plots use event-to-voxel occupancy,

$$\eta := \frac{N_{\mathrm{init}}}{n_x n_y n_z},$$

as the horizontal variable. At fixed beam energy and sweep count, these plots compare the proposal fraction $\mathbb{P}_{\widehat{p}}((1,0))$ and the absolute uncalibrated peak-depth offsets against η. Event-multiplicity comparisons also use uncalibrated peak-depth offsets.

The first offset compares the reconstructed peak with the fast-neutron truth peak. The second compares it with the geometry-only peak. Neither offset uses the truth-to-truth calibration coefficients. Within-energy comparisons hold beam energy fixed and vary sweep count, reconstruction grid, or data provenance. Across-energy comparisons hold the cohort definition fixed and compare one derived summary per beam energy.

4.5.7 Repository and data availability

The public SOE reconstruction core, configuration files, and input datasets are available at https://github.com/haljoumaa/soe-fn. This repository contains the code and data needed to reproduce the reconstruction runs described in this thesis. Offline analysis scripts are maintained separately from the public reconstruction repository and can be made available by the author upon reasonable request.

Chapter 5

Results

5.1 Truth

Before reconstruction is evaluated, the truth profiles must establish whether fast-neutron production contains a usable signal for tracking the operational reference depth p_{ref} across beam energy.

5.1.1 Feature calibration

Among the four with-scatter fast-neutron truth features, peak depth gives the lowest calibration residual against p_{ref} and remains monotone across all analysed energies. R_{50} is retained as the secondary calibrated diagnostic because it also remains extractable after reconstruction. R_{20} passes the truth-level screening rule, but is not propagated to reconstructed range prediction because the reconstructed distal tail does not consistently cross 20% of the peak within the analysed field of view. R_{90} is rejected at truth level because its truth-profile response is not monotone with beam energy.

Table 5.1: Truth-to-truth calibration metrics for fast-neutron features against p_{ref}. Metrics are computed over all eight analysed beam energies.

Feature	Pearson r	RMSE (cm)	MaxAE (cm)	Monotone
Peak depth	0.9942	0.240	0.581	Yes
R_{20}	0.9891	0.328	0.591	Yes
R_{50}	0.9847	0.388	0.669	Yes
R_{90}	0.9660	0.577	1.136	No

The locked calibration coefficients propagated to the reconstruction analysis are

$$\hat{p}^{(\mathrm{peak})}(E_b) = 3.405 + 1.410\, d_{\mathrm{truth}}^{(\mathrm{peak})}(E_b), \tag{5.1}$$

$$\hat{p}^{(R_{50})}(E_b) = -14.751 + 2.287\, d_{\mathrm{truth}}^{(R_{50})}(E_b), \tag{5.2}$$

with all depths in centimetres. These two maps are the only calibration maps used for reconstructed range estimates below.

5.1.2 Truth-profile morphology

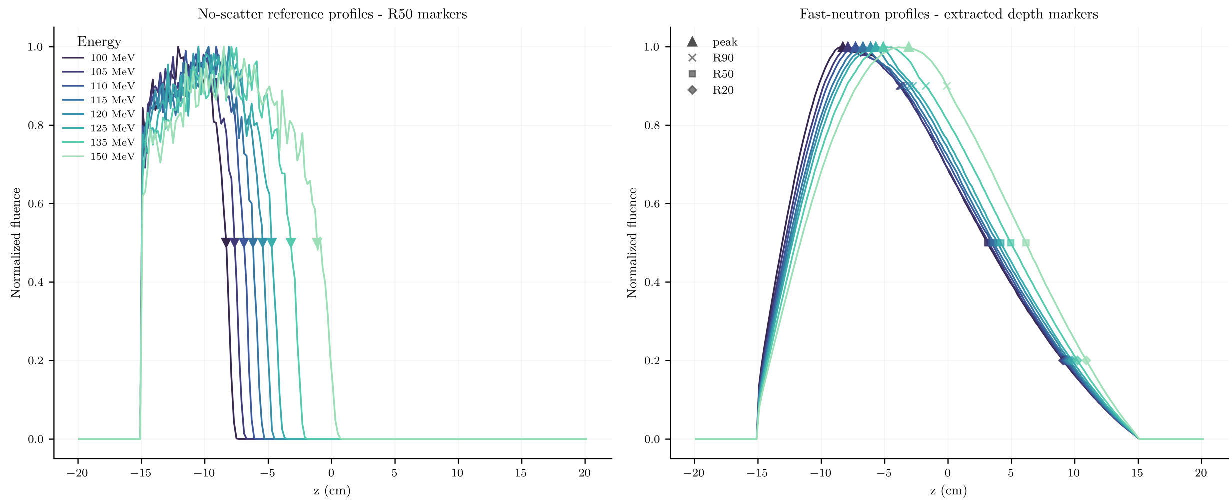


Figure 5.1: Peak-normalised truth profiles across energy. Left: `n-events` reference profiles with p_{ref} R_{50} markers. Right: fast-neutron `all-with-scatters` profiles with peak-depth, R_{90}, R_{50}, and R_{20} markers.

The operational reference profiles show a sharp distal falloff that shifts systematically with beam energy. The fast-neutron truth profiles preserve the same energy ordering, but their feature markers lie on a broader distal region.

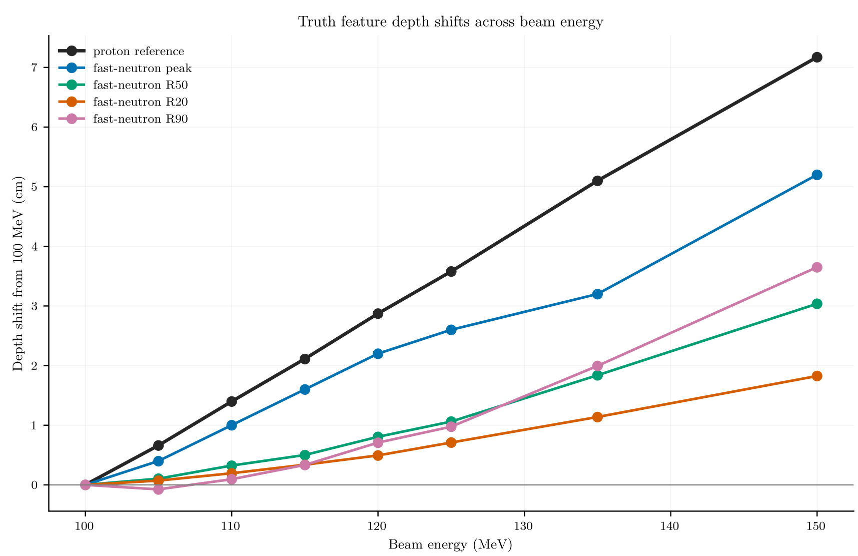


Figure 5.2: Truth-feature shifts across beam energy, referenced to 100 MeV.

From 100 MeV to 150 MeV, p_{ref} shifts by 7.17 cm. Over the same interval, the with-scatter truth peak depth shifts by 5.20 cm, giving $C_{\text{truth}}^{(\text{peak})} = 0.73$. The truth-level peak therefore preserves the energy ordering while compressing the operational reference-depth shift.

5.2 Geometry

The geometry layer shows broad axial accessibility with weak energy dependence before SOE dynamics are applied.

5.2.1 Eventwise axial-information statistics

Table 5.2: Eventwise geometry-layer statistics over the eight analysed energies.

E_b (MeV)	N_{area}	Median width (cm)	Width fraction	$> 50\%$ VOI (%)	Median TV	TV < 0.10 (%)	Median $\lvert u \cdot b \rvert$	p_{ref} (cm)
100	4244	25.26	0.842	79.4	0.0957	52.2	0.441	−8.31
105	3862	25.35	0.845	80.7	0.0953	52.3	0.430	−7.65
110	4058	25.10	0.837	80.5	0.0951	52.4	0.426	−6.91
115	4262	25.43	0.848	80.2	0.0979	51.3	0.426	−6.19
120	4305	25.10	0.837	79.7	0.0975	51.5	0.426	−5.43
125	4661	25.28	0.843	81.4	0.1003	49.7	0.442	−4.73
135	4831	25.38	0.846	80.6	0.0997	50.1	0.429	−3.21
150	7104	25.87	0.862	82.3	0.0972	51.4	0.408	−1.13

The median axial support width remains between 25.10 cm and 25.87 cm, corresponding to 0.837–0.862 of the 30 cm VOI. Across all energies, about four fifths of the events span more than half of the axial window. At the event level, cone geometry is broad rather than depth-localised.

Within those broad supports, the eventwise axial accessibility laws remain close to their footprint-uniform references. Median TV distance remains near 0.10, and about half of the events have TV distance below 0.10. Median beam-axis alignment stays between 0.408 and 0.441, with no monotone energy trend.

5.2.2 Aggregate cross-energy axial accessibility

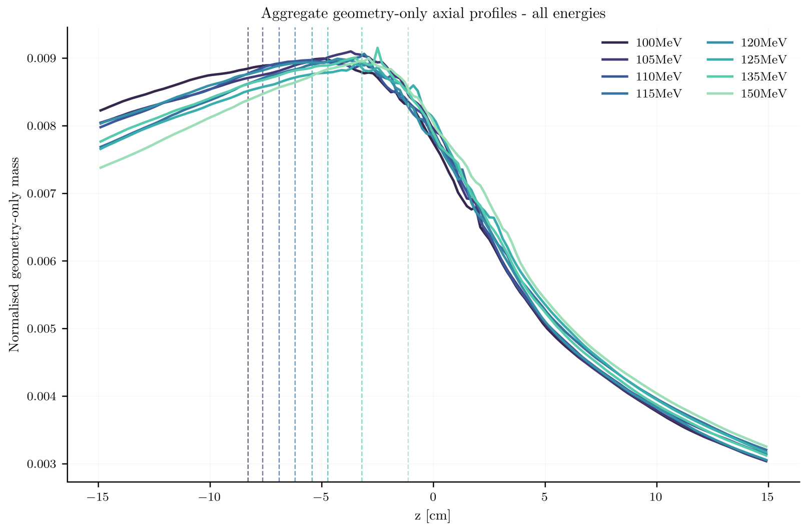


Figure 5.3: Cross-energy geometry aggregate axial accessibility profiles with p_{ref} R_{50} dashed lines.

The aggregate axial-accessibility profiles remain nearly energy-invariant. Each profile has a broad maximum between $z \approx -7$ and -3 cm, while p_{ref} shifts from -8.31 cm to -1.13 cm over the analysed energies.

Table 5.3: Cross-energy peak-depth span compression at the geometry layer over the eight analysed energies.

Truth peak-depth span (cm)	Geometry peak-depth span (cm)	$C_{\mathrm{geom}}/C_{\mathrm{truth}}^{(\mathrm{peak})}$	p_{ref} span (cm)	C_{geom}
5.20	3.00	0.577	7.17	0.418

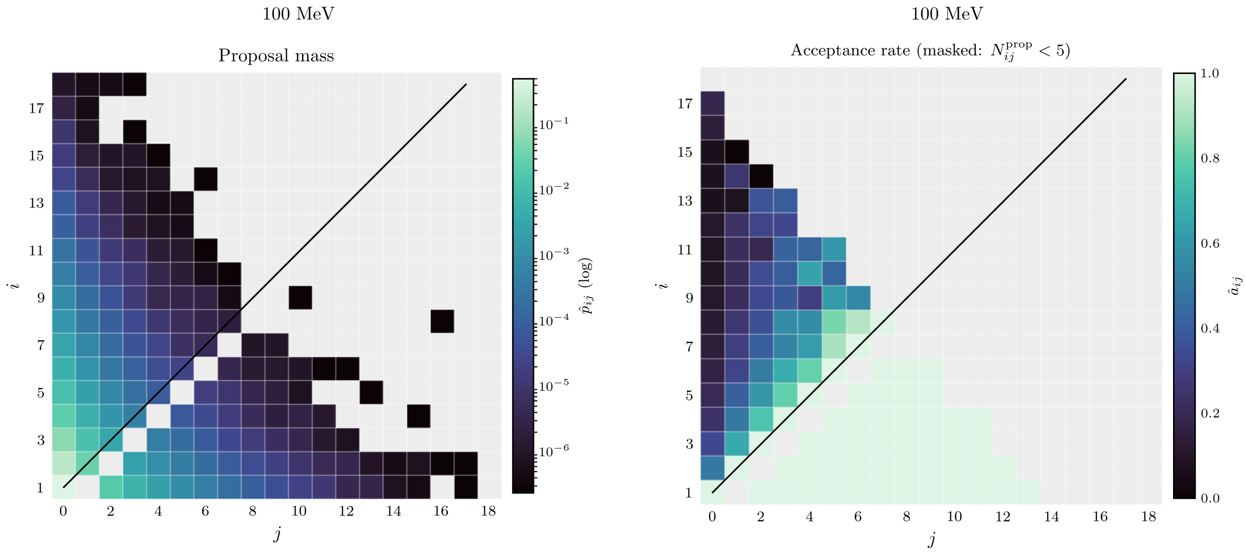
Over all eight analysed energies, the geometry-only peak-depth span is 3.00 cm. This corresponds to 0.577 of the fast-neutron truth peak-depth span and 0.418 of the reference-depth span.

5.3 Dynamics

The dynamics layer tests how the SOE transition kernel reshapes the admitted event information during sampling.

5.3.1 Low-count transition structure

In the anchor run, empirical proposal mass concentrates at low departure and arrival occupancies. The largest cell is $(1, 0)$, corresponding to proposals from singleton departure voxels into empty arrival voxels.



(a) Proposal mass over low-count pairs.

(b) Conditional acceptance rate.

Figure 5.4: Empirical low-count transition structure in the anchor run.

The realised acceptance pattern follows the threshold $i \leq j+1$. Cells on or below that threshold are accepted at, or near, probability one; cells with $i > j+1$ have lower conditional acceptance. Within the displayed low-count range, accepted motion is dominated by singleton-to-empty and other sparse transitions.

5.3.2 Sweep-depth response

For the single-seed 100 MeV, $10 \times 15 \times 150$, S0 sweep-depth cohort, the proposal regime remains stable across the sweep ladder.

Table 5.4: Transition statistics versus sweep depth.

Sweeps	$\mathbb{P}_{\widehat{p}}(j=0)$	$\mathbb{P}_{\widehat{p}}(i=1)$	$\mathbb{P}_{\widehat{p}}((1,0))$	$\mathbb{P}_{\widehat{p}}(j<i-1)$
500	0.7976	0.6245	0.5017	0.3220
1000	0.7972	0.6248	0.5018	0.3216
2000	0.7973	0.6254	0.5024	0.3211
5000	0.7972	0.6257	0.5026	0.3207
10000	0.7972	0.6256	0.5024	0.3210

All four transition summaries vary by less than 0.0015 in absolute value. The $(1,0)$ cell remains the dominant proposal cell at every sweep depth.

Table 5.5: Single-seed reconstructed features and absolute calibrated range errors versus sweep depth.

Sweeps	Peak depth (cm)	R_{50} depth (cm)	Abs. peak-range error (cm)	Abs. R_{50}-range error (cm)
500	-2.90	4.113	7.621	2.961
1000	-4.50	4.118	5.364	2.972
2000	-4.10	4.142	5.929	3.026
5000	-4.10	4.262	5.929	3.301
10000	-3.90	4.247	6.211	3.266

The smallest absolute peak-based range error occurs at 1000 sweeps. Beyond that point, the absolute peak-range error remains between 5.929 cm and 6.211 cm. The absolute R_{50}-range error stays near 3 cm, with the largest values at 5000 and 10 000 sweeps.

5.3.3 Sparsity and reconstruction response

Transition statistics versus peak-depth position

Within each fixed-energy grid-comparison set defined in section 4.5.1, higher event-to-voxel occupancy η reduces $\mathbb{P}_{\widehat{p}}((1,0))$. The event-multiplicity control extends the same transition-statistic trend beyond the occupancy range of the three native event sets.

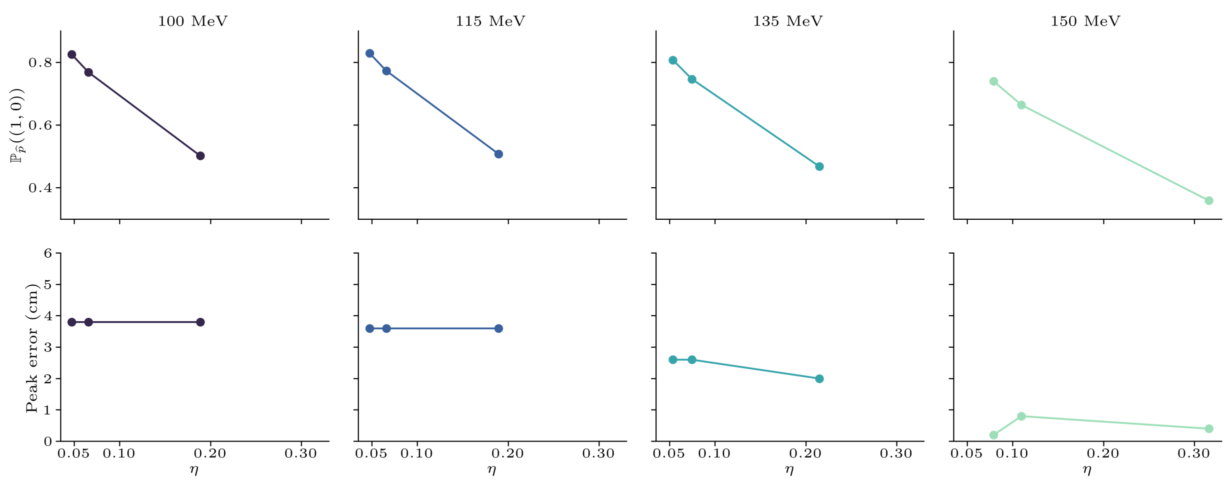


Figure 5.5: Within-energy response of $\mathbb{P}_{\widehat{p}}((1,0))$ and the absolute uncalibrated peak-depth offset from fast-neutron truth as functions of event-to-voxel occupancy η.

The peak-depth offset does not show the same response. It is nearly flat at 100 MeV and 115 MeV. At 135 MeV and 150 MeV, it changes weakly and without a common monotone direction.

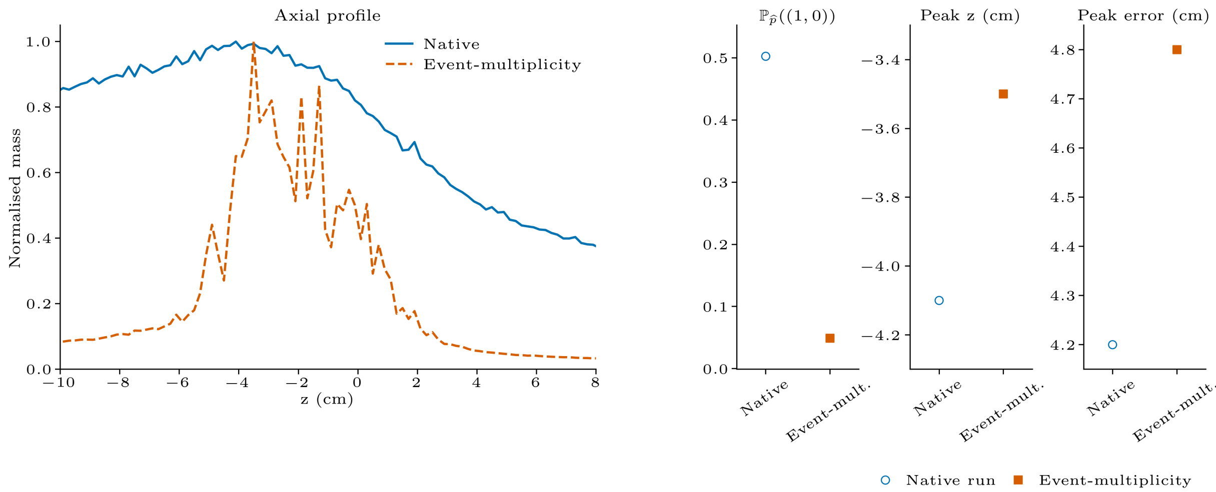


Figure 5.6: Event-multiplicity control at fixed native geometry source.

The event-multiplicity run contains 49 950 admitted events after preprocessing. This higher event-to-voxel occupancy sharply reduces $\mathbb{P}_{\hat{p}}((1,0))$. The reconstructed profile and the absolute uncalibrated peak-depth offset remain comparable to the native runs.

Peak-depth tracking across energy

Across the four grid-comparison energies, the reported reconstructed peak depth is the arithmetic mean of the deduplicated peak depths from the three retained reconstruction grids. This cross-grid diagnostic is separate from the seed-averaged native-grid range-verification metrics reported in section 5.4. The reconstructed peak depth spans 1.80 cm. Over the same energies, the geometry-only peak depth spans 2.40 cm, while the with-scatter truth peak depth spans 5.20 cm. This four-energy geometry span differs from the all-energy geometry-layer span of 3.00 cm because the all-energy calculation includes the 110 MeV geometry peak at −5.50 cm.

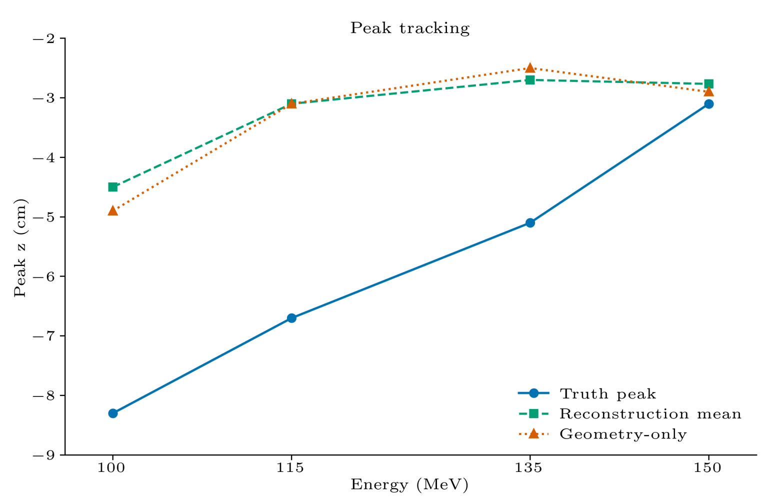


Figure 5.7: Peak-depth tracking across the four grid-comparison energies.

At 100 MeV, 115 MeV, and 135 MeV, the reconstructed peak depth lies closer to the geometry-only peak depth than to the with-scatter truth peak depth. At 150 MeV, all three peak depths are within 0.33 cm of each other.

Table 5.6: Reconstructed, geometry-only, and truth peak-depth positions over the four grid-comparison energies.

E (MeV)	$z^{\text{rec}}_{\text{peak}}$ (cm)	$z^{\text{geom}}_{\text{peak}}$ (cm)	$z^{\text{truth}}_{\text{peak}}$ (cm)	$\lvert z^{\text{rec}}_{\text{peak}} - z^{\text{geom}}_{\text{peak}}\rvert$ (cm)	$\lvert z^{\text{rec}}_{\text{peak}} - z^{\text{truth}}_{\text{peak}}\rvert$ (cm)
100	-4.50	-4.90	-8.30	0.40	3.80
115	-3.10	-3.10	-6.70	0.00	3.60
135	-2.70	-2.50	-5.10	0.20	2.40
150	-2.77	-2.90	-3.10	0.13	0.33
span over four energies	1.80	2.40	5.20		

At all four grid-comparison energies, the reconstructed peak depth is closer to the geometry-only peak depth than to the with-scatter truth peak depth. Its cross-energy span is also closer to the geometry-only span than to the with-scatter truth span.

5.4 Range

5.4.1 Axial-profile and spatial comparison

The axial mismatch is largest at low beam energy. The axial comparison uses the with-scatter fast-neutron product-depth profile.. From 100 MeV to 115 MeV, the reconstructed peak depth remains distal to the with-scatter truth peak depth, and the reconstructed distal falloff is broad. From 120 MeV to 150 MeV, the profiles narrow and the peak-depth displacement decreases.

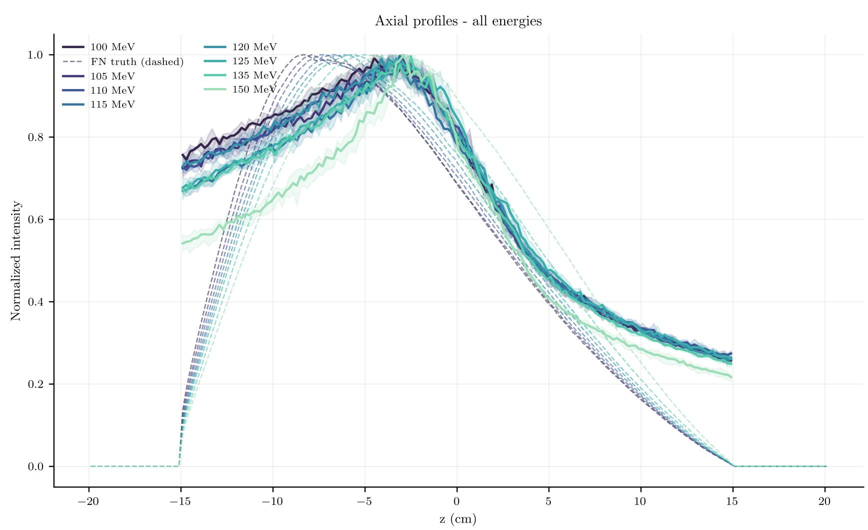
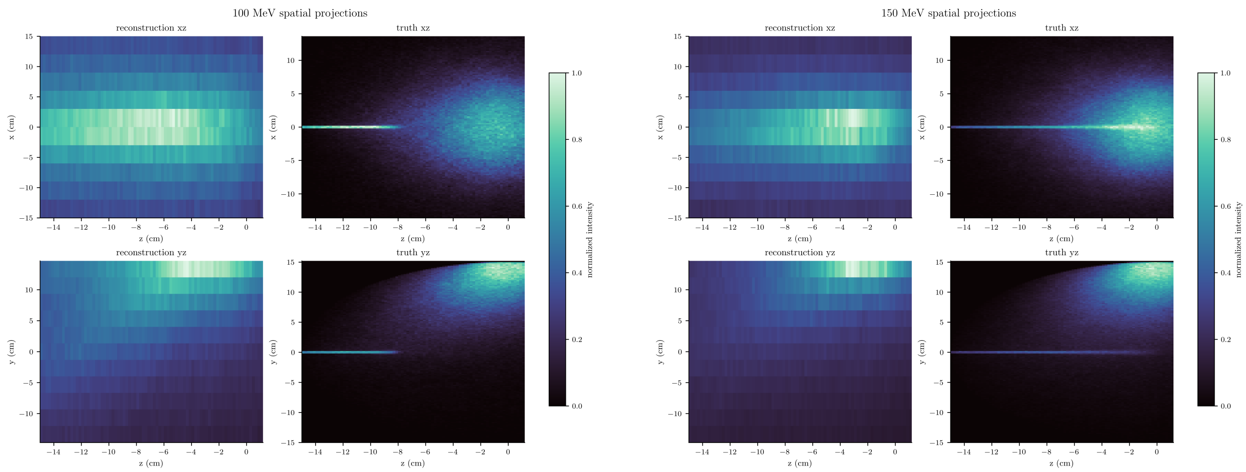


Figure 5.8: Peak-normalised reconstructed axial profiles compared with the PHITS with-scatter neutron-production tally.

The spatial projections show the same low–high energy contrast. The truth panels use the with-scatter fast-neutron truth volume, projected onto the same coordinate planes as the reconstruction. Each panel is normalised by its own maximum and cropped to the overlapping coordinate support. The 100 MeV reconstruction remains diffuse, whereas the 150 MeV reconstruction concentrates into a more compact region.



(a) 100 MeV.

(b) 150 MeV.

Figure 5.9: Spatial projections at 100 MeV and 150 MeV.

5.4.2 Energy-resolved range predictions

For peak depth, the signed seed-mean range error $\bar{e}^{(\mathrm{peak})}$ is positive at every energy. It ranges from 0.54 cm at 150 MeV to 6.04 cm at 110 MeV. The peak-depth seed standard deviation remains below 0.71 cm. The signed error decreases toward 150 MeV, but not monotonically. For R_{50}, $\bar{e}^{(R_{50})}$ is smaller in magnitude than the peak-depth error from 100 MeV to 125 MeV. It changes sign at 135 MeV and reaches −5.48 cm at 150 MeV.

Table 5.7: Energy-resolved calibrated range predictions. Predictions are seed mean ± one standard deviation.

Beam energy (MeV)	Reference p_{ref} (cm)	Peak-depth prediction(cm)	Peak-depth signed range error (cm)	R_{50} prediction (cm)	R_{50} signed range error (cm)
100	−8.31	-2.85 ± 0.16	5.46	-5.17 ± 0.21	3.14
105	−7.65	-1.81 ± 0.56	5.83	-5.28 ± 0.49	2.36
110	−6.91	-0.87 ± 0.16	6.04	-5.12 ± 0.44	1.79
115	−6.19	-0.87 ± 0.16	5.32	-5.06 ± 0.30	1.13
120	−5.43	-1.91 ± 0.43	3.53	-5.11 ± 0.47	0.33
125	−4.73	-0.69 ± 0.28	4.04	-4.40 ± 0.63	0.33
135	−3.21	-0.87 ± 0.71	2.33	-5.54 ± 0.21	−2.33
150	−1.13	-0.59 ± 0.16	0.54	-6.61 ± 0.50	−5.48

The R_{50} calibration gives smaller errors from 100 MeV to 125 MeV. Its signed error changes direction at 135 MeV and reaches −5.48 cm at 150 MeV.

5.4.3 Aggregate range performance

Both calibrated features compress the reference-depth dynamic range. The predicted spans $D^{(f)}_{\mathrm{pred}}$, computed from the seed-mean calibrated estimates, are 2.26 cm for peak depth and 2.21 cm for R_{50}. The operational reference span $D_{p_{\mathrm{ref}}}$ is 7.17 cm, giving $C^{(\mathrm{peak})}_{\mathrm{span}} = 0.315$ and $C^{(R_{50})}_{\mathrm{span}} = 0.308$.

The R_{50} feature has lower aggregate error than peak depth by both $\mathrm{RMSE}_{\mathrm{rec}}$, computed from signed seed-mean errors, and $\mathrm{MAE}_{\mathrm{rec}}$, computed from per-energy mean absolute seed errors. Neither calibrated feature is monotone in beam energy. Seed variability is smaller than the aggregate errors, with mean seed standard deviations of 0.33 cm for peak depth and 0.41 cm for R_{50}.

Table 5.8: Aggregate calibrated range-prediction performance across all eight analysed energies. The span ratio is predicted span divided by reference span.

Feature	RMSE (cm)	MAE (cm)	$\mathrm{MaxMeanAE}_{\mathrm{rec}}$ (cm)	Monotone	Pred. span (cm)	Ref. span (cm)	Span ratio	Mean seed SD (cm)	Max seed SD (cm)
Peak depth	4.51	4.14	6.04	No	2.26	7.17	0.315	0.33	0.71
R_{50}	2.64	2.15	5.48	No	2.21	7.17	0.308	0.41	0.63

5.5 Compute

The timed reconstructions were dominated by the Markov-chain core. For the 150 MeV run, sampled admission retained $N = 7104$ events and the chain attempted 7,104,000 single-event updates. The chain core took 813.9 s, corresponding to 8728 attempted updates per second, 0.115 ms per attempted update, and 0.814 s per sweep. The full reconstruction completed in 818.5 s (13.6 min), so the chain core accounted for 99.4% of the measured wall time.

Table 5.9: Measured reconstruction wall time for the timed range-verification runs. The chain core denotes the Metropolis–Hastings update loop.

Energy	N	Attempted steps	Preprocessing (s)	Chain core (s)	Retained mean (s)	Total (s)
100 MeV	4244	4,244,000	1.2	237.8	0.024	239.4
150 MeV	7104	7,104,000	3.4	813.9	0.054	818.5

The second timed run gives a limited scaling check rather than a fitted scaling law. With the number of sweeps fixed at 1000, the attempted-step count was $1000N$. Increasing the surviving event count from 4244 to 7104 increased the total wall time from 239.4 s to 818.5 s.

Chapter 6

Discussion

The central result is that the analysed reconstructions do not preserve the range signal. At truth level, peak depth is monotone over the analysed energies and gives the lowest truth-to-truth calibration residual. After reconstruction, the calibrated features compress the reference range span, lose monotonicity, and produce centimetre-scale range errors that exceed the observed seed variability.

This negative result is conditional. It applies to the simulated lateral detector arrangement, the admitted cone-event data, the SOE reconstruction regime, and the feature-based range-estimation pipeline used in this work. The truth profiles do contain range-dependent axial information, and different geometries, likelihood models, priors, proposal rules, or estimators could change how that signal is represented. The narrower conclusion is that, in the tested configuration, the range-relevant signal is not retained in the reconstructed image.

The discussion therefore explains how a truth-level surrogate with measurable range dependence becomes a reconstructed feature set with compressed span and unreliable ordering.

6.1 Geometric compression of the axial signal

Range verification with secondary particles requires a reconstructed depth profile that remains correlated with the proton range. This premise underlies both fast-neutron production-distribution analyses and prompt-gamma systems where range estimation relies on the emission profile rather than on total detected yield [16, 37]. The geometry result therefore addresses the first necessary condition for the pipeline: whether the recorded cone data encode enough beam-axis information before SOE acts on them.

In the simulated setup, the proton beam propagates in the $+z$ direction while the detector view is transverse. Each recorded event constrains the emission origin to a cone surface, so axial localisation must arise from how those cone surfaces intersect the reconstruction volume across many events. Detector arrangement and element orientation affect fast-neutron production-distribution quality and range-shift sensitivity [17], so the relevant question here is not the general feasibility of fast-neutron range verification, but whether this configuration preserves the axial information needed for it.

The geometry-only results in table 5.2 show that the admitted cones contain axial information, but mostly as broad, weakly energy-dependent accessibility rather than narrow depth localisation. The median eventwise axial support covers 0.837–0.862 of the 30 cm axial window, and about four fifths of the events span more than half of that window. Within those broad supports, the eventwise axial accessibility laws remain close to uniform, with median TV distance near 0.10. These events therefore do not impose narrow depth constraints along the beam axis; they admit broad axial regions that differ only weakly from one energy to another.

The aggregate accessibility profiles confirm the same pattern at the cross-energy level (table 5.6). Their broad maxima remain concentrated around $z \approx -7$ to -3 cm, while the proton reference range shifts from -8.31 cm to -1.13 cm. The geometry-only peak-depth span is 3.00 cm, compared with 5.20 cm for the fast-neutron truth peak and 7.17 cm for the proton reference range. Before SOE redistributes any event origins, the cone geometry has already compressed the energy-dependent axial signal.

This compression is configuration-specific. Other detector designs can preserve range information more effectively, and lateral arrangements are not intrinsically unusable; hybrid studies have found high sensitivity for lateral orientations under other measurement configurations [38]. In the present PHITS-based simulation, however, the admitted event cones encode the fast-neutron depth signal with limited beam-axis contrast. SOE can redistribute event origins on the admitted cone surfaces, but it cannot recover a strongly range-correlated axial profile from cone data that already encode the beam-axis signal weakly.

6.2 Dynamics within a weakly informative state space

SOE can only redistribute event origins within the measured cone constraints. It does not directly observe the fast-neutron production distribution; it seeks to infer that distribution by iteratively relocating representative event origins on the measured cone surfaces [20, 21]. A reconstruction can therefore remain algorithmically active while sampling a weakly informative state space. Chain movement alone is not evidence that the fast-neutron depth profile has been recovered.

The transition diagnostics support this distinction. In the anchor run, proposal mass concentrates at low departure and arrival occupancies, with the $(1, 0)$ transition as the dominant cell (fig. 5.4). This is not merely a high acceptance regime; it is a sparse reassignment regime in which many accepted moves relocate a singleton event origin into an empty voxel. The resulting chain can continue to move without generating a sharper or more truth-like axial profile. This interpretation is consistent with the broader SOE literature, where continued iteration can coexist with increased variance, bias, or degradation after an equilibrium regime has been reached [20, 21].

Increasing the number of sweeps did not change that regime. The empty-arrival probability, singleton-departure probability, $(1, 0)$ proposal fraction, and downhill-transition probability all varied by less than 0.0015 across the sweep ladder (table 5.4). The reconstructed range features did not improve monotonically with additional sweeps either (table 5.5): the smallest peak-based range error occurred at 1000 sweeps, while longer chains remained several centimetres from the calibrated reference. More sweeps therefore sampled the same low-information transition regime for longer without recovering the missing axial signal.

The sparsity controls separate transition sparsity from axial recovery. Within the fixed-energy grid-comparison sets, increasing event-to-voxel occupancy reduced the dominance of $(1, 0)$ transitions (fig. 5.5). The event-multiplicity control extended the same transition-statistic trend at fixed native geometry source (fig. 5.6). The reconstructed peak-depth offset did not show the same response: it remained weakly varying, non-monotone, or comparable to the native runs. Sparse occupancy therefore contributes to the transition regime, but reducing the $(1, 0)$ proposal fraction does not by itself restore range-relevant axial localisation.

The reconstructed peak positions instead remained closer to the geometry-only profile than to the fast-neutron truth profile. The four-energy grid-comparison diagnostic shows this point directly: at all four comparison energies, the reconstructed peak depth lies closer to the geometry-only peak depth than to the with-scatter truth peak depth (table 5.6). The all-energy span accounting gives the same conclusion at the fixed range-verification scale. Over the eight analysed energies, the with-scatter truth peak-depth span is 5.20 cm, while the geometry-only peak-depth span is 3.00 cm. The calibrated peak-depth prediction span is 2.26 cm; using the calibration slope in eq. (5.1), this corresponds to a reconstructed peak-depth span of 1.60 cm. On this peak-depth scale, geometry accounts for 2.20 cm of the 3.60 cm truth-to-reconstruction span loss, or about 61%. The SOE reconstruction accounts for the remaining 1.40 cm, or about 39%. SOE therefore did not transform the compressed geometry-level axial signal into the truth-level depth signal. It compressed the span further.

This does not show that SOE is generally unsuitable for emission reconstruction. Prior studies report successful behaviour in better conditioned or better modelled settings, and also report degraded behaviour when statistics, angular coverage, resolution, or cone accuracy are unfavourable [20, 25]. In the present configuration, SOE stabilises around a low-information axial solution. Without stronger constraints, priors, or a more axially informative event set, the reconstruction dynamics do not recover the depth signal required for range verification.

6.3 Range-estimation error as upstream signal loss

Profile-based range verification depends on extracting a range-correlated feature from a measured or reconstructed axial profile, typically in the distal region where range shifts are expressed most directly [4, 39]. The analysed fast-neutron truth profiles satisfy this premise: neutron-production depth distributions correlate with the primary proton range [16]. In this work, truth-level fast-neutron peak depth retained an energy-ordered range signal, gave the lowest truth-to-truth calibration residual, and retained 0.73 of the reference range span in the truth-to-truth calibration results.

The downstream range-estimation error therefore does not imply that fast-neutron truth profiles lack range information. It shows that the reconstructed profiles did not preserve enough of it. This distinction matters because range estimation acts only on the feature supplied to it. Compton-camera studies report the same vulnerability: reconstructed secondary-emission profiles can have displaced maxima or degraded fall-off regions, and range evaluation may then depend on whichever profile feature remains most reliably correlated with dose or range [37, 40]. PET-based verification applies the same logic, extracting range markers from one-dimensional activity distributions rather than from the full emission field [41, 42]. In this study, both calibrated reconstructed features did not satisfy that requirement: peak depth and R_{50} compressed the reference range span to 0.315 and 0.308, respectively, and neither remained monotone in beam energy.

Calibration did not create this loss. The locked truth-to-truth calibration mapped fast-neutron truth features to proton range with sub-centimetre residuals. The reconstructed features entered that same calibration after axial displacement, broadening, and span compression. A linear calibration can correct a stable offset or scale difference; it cannot restore energy ordering or usable cross-energy separation once the reconstructed feature has lost them. The centimetre-scale range errors are the downstream expression of that upstream loss: peak depth reached an RMSE of 4.51 cm, and R_{50} reached 2.64 cm in the reconstructed range-error summaries. Both values are far above the millimetre-scale range precision sought in particle-therapy range-verification studies [4, 7].

The range-estimation stage therefore exposed the limitation rather than causing it. Truth-level fast-neutron peak depth carried range-relevant information, but the geometry and SOE reconstruction regime delivered features whose axial span, ordering, and distal structure no longer matched the truth profile. The range prediction became unreliable because the feature was already distorted before calibration was applied.

6.4 Computational practicality and scaling

The measured runtime is adequate for the offline role assigned to the pipeline in this study. The slowest timed reconstruction completed in 13.6 min on a consumer laptop, without parallel execution or compiled extensions. This cost allowed the reported energy-dependent reconstructions to be generated and compared within an exploratory feasibility workflow. It is not evidence of real-time or online adaptive feasibility. The present implementation should therefore be interpreted as adequate for offline analysis, not as a candidate for online adaptive use without further acceleration.

The dominant cost follows directly from the single-event Metropolis–Hastings structure. One step proposes one update for one selected surviving event. A sweep is the derived accounting unit of N such steps, where N is the number of surviving events. Thus a run reported as S sweeps attempts SN single-event updates, while a run specified directly by K steps attempts K updates. At fixed sweep count S, the leading algorithmic control variable is therefore N. The two timed reconstructions are consistent with this dependence in the sense that the larger admitted event set required more attempted updates and more wall time. They should not be read as a clean asymptotic fit, because the proposal cost also depends on event geometry and clipped-cone support. The appropriate scaling statement is therefore that the realised wall time scales with the attempted single-event update count times the average event-dependent proposal cost.

The reported timing is the result of targeted implementation work rather than an unoptimised dense-array prototype. Development profiling showed that an unoptimised dense-array implementation spent most of its per-step time on defensive copying and validation of the occupancy array, while the proposal, geometry, and acceptance calculation accounted for only a small fraction of the measured step cost. Those costs were implementation overheads, not consequences of the SOE target law. The final implementation therefore uses a trusted internal update path, event-level geometry caching, retained-mean accumulation during the chain, and a direct surface sampler for the bounded-box proposal. These changes preserve the

Metropolis–Hastings transition law but move the bottleneck away from avoidable dense-array bookkeeping.

After these changes, the remaining cost is mainly proposal-side geometry work. Each accepted or rejected move still requires sampling a new point on the clipped cone surface and evaluating the corresponding occupancy change. This cost is local in the selected event, but it is paid once per attempted single-event update: K times for a run specified in steps, or SN times for a run specified as S sweeps over N surviving events. Dense voxel-grid objects still enter through the occupancy field and retained mean, so memory use retains an $O(|V|)$ component. The refactors removed the more damaging $O(\text{steps}\,|V|)$ dense-history behaviour from the standard path, but they do not make the method independent of grid size.

The practical limitation is therefore clear. For the grid and event counts used here, a single-process Python implementation is fast enough for offline feasibility experiments. Larger event sets, larger voxel grids, heterogeneous phantoms, or repeated uncertainty studies would increase cost through N, $|V|$, proposal complexity, or repeated runs. Further reductions would most naturally come from compiled proposal kernels, parallel execution across independent proposal evaluations, or GPU/array-level treatment of geometry operations. Those changes would affect runtime engineering rather than the statistical model.

Chapter 7

Conclusion

7.1 Summary of findings

This thesis formalized and implemented an SOE reconstruction pipeline for simulated fast-neutron cone data and evaluated whether the reconstructed axial profiles could serve as range-verification surrogates. Under the detector geometry, event statistics, voxelisation, and feature-extraction rules used here, the answer is negative. The truth profiles contain range-relevant information, but the reconstructed features do not preserve enough of that information to give reliable range estimates.

The loss does not originate at truth level. Fast-neutron peak depth remains monotone across the analysed energies, gives the lowest truth-to-truth calibration residual, and retains 0.73 of the reference range span. The first major compression appears before reconstruction: the geometry-only peak-depth span retains 0.577 of the fast-neutron truth span and 0.418 of the reference range span. The admitted cone geometry therefore carries an axial signal, but with reduced beam-axis contrast.

SOE does not recover the compressed signal. Across the grid-comparison energies, the reconstructed peak depth remains closer to the geometry-only peak than to the fast-neutron truth peak, and its cross-energy span is smaller than the geometry-only span. The calibrated reconstructed features then compress the reference range span to 0.315 for peak depth and 0.308 for R_{50}. Both features also lose monotonicity and produce centimetre-scale range errors.

The range-estimation error is therefore the downstream expression of upstream signal loss. The locked truth-to-truth calibration maps fast-neutron truth features to the reference range with sub-centimetre residuals, but the reconstructed features enter that calibration after axial displacement, broadening, and span compression. Calibration exposes this distortion; it does not cause it. The implementation is computationally adequate for offline feasibility analysis, but the tested reconstruction target and measurement configuration do not retain the axial information required for reliable range-verification use.

7.2 Future work

Future work should not begin by adding complexity to the reconstruction model alone. The results already indicate that most peak-depth span loss occurs before SOE dynamics, at the geometry level. The next step is therefore not to ask whether SOE can be made more complex, but to identify which part of the geometry, event-admission, and feature-extraction chain limits axial leverage, and whether targeted changes recover energy separation before the reconstruction law is modified.

The detector geometry is the first object to test because it sets the axial leverage available in the cone data before the reconstruction begins. A geometry that produces cones with weak axial discrimination cannot be repaired by the sampler unless the missing information is still present in another form. Future work should therefore test alternative detector placements, angular coverages, detector distances, layer spacings, and source–detector configurations.

Sparse valid-event statistics should also be treated as a physical limitation, not only as a simulation inconvenience. Realistic range-verification conditions are expected to involve limited usable event counts, event-selection losses, and scatter-induced ambiguity [25]. These limitations are especially relevant for SOE because low-count regimes can make the reconstruction more sensitive to event statistics and model constraints [25]. A useful next study would compare three event cohorts: ideal no-scatter events, controlled-scatter events, and full scatter-aware events. The no-scatter cohort would isolate the geometric information content of the cone model. The full-scatter cohort would test the reconstruction under a more realistic but less mechanistically transparent regime.

A further limitation is the occupancy-only target law in the SOE formulation. The acceptance rule rewards changes in voxel occupancy but does not explicitly encode event reliability, cone geometry, or local spatial coherence. This may be too weak in sparse regimes, where individual event assignments carry limited and uneven axial information. Future work should therefore test a geometry-aware or neighbourhood-regularised SOE law in which acceptance depends not only on occupancy change, but also on neighbourhood consistency, event-specific geometric leverage, and cone uncertainty or event quality. A Gibbs- or Markov-random-field-inspired extension [43] would be a controlled way to test whether a stronger target law can recover residual axial structure that remains after the geometry and event-selection stages have already compressed the signal.

The same limitation can be formalised as an information-channel problem. The physical proton interaction field is filtered through fast-neutron production, neutron transport, detector response, event selection, cone construction, stochastic sampling, feature extraction, and calibration. Future work should quantify the loss at each stage, with particular emphasis on the pre-SOE geometry and event-selection stages identified here as the dominant compression point. This analysis should remain diagnostic unless information-theoretic quantities such as mutual information or channel capacity are actually computed [44]. Without such computation, the channel view should be used as a disciplined framework for separating measurement loss, reconstruction loss, and feature-loss.

The work is simulation-based, so the next validation step should first remain within simulation. Future studies should test whether the observed geometry-level compression persists under more realistic detector response, event-selection uncertainty, phantom heterogeneity, and independent simulated datasets. PHITS remains an appropriate basis for such simulation studies [26], but agreement within one simulation configuration cannot establish robustness across detector designs or measurement conditions. Measured detector data then form the later validation boundary, once the pipeline has first shown stable behaviour under controlled simulation extensions.

Prompt-gamma and fast-neutron fusion remains a natural NOVO-aligned extension [45], because the two particle species carry complementary range-related information along the beam path [17, 46]. That complementarity does not remove the need to validate the fast-neutron reconstruction channel on its own. Future work should therefore first establish which axial information is retained and lost in the fast-neutron SOE pipeline, then test whether prompt-gamma information adds range sensitivity under the same range-surrogate metrics. In this order, fusion becomes a controlled multimodal extension rather than a way to mask the information limit identified in the fast-neutron channel.

7.3 Concluding remarks

Under the studied conditions, SOE is not supported as a reliable range-verification method for sparse, information-limited fast-neutron data. The main value of this result is that it separates a geometry-level information limit from a reconstruction-dynamics limit. This separation identifies what future detector layouts, event-selection strategies, or reconstruction laws would need to overcome before SOE can be used as a range-verification method in this setting.

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Appendix A

Derivations and Proofs

A.1 Surface-area measure on feasible cone surfaces

With the notation of section 3.1, let H^2 denote the two-dimensional Hausdorff measure in $\mathbb{R}^3$ [47]. Since C_e and Ω are Borel subsets of $\mathbb{R}^3$, the feasible surface $\Sigma_e = C_e \cap \Omega$ is Borel. The event-wise surface-area measure is the restriction of H^2 to the feasible surface:

$$\mu_e(A) := H^2(A \cap \Sigma_e), \qquad A \subset \mathbb{R}^3 \text{ Borel.}$$

Thus $\mu_e = H^2 \restriction_{\Sigma_e}$, regarded as a Borel measure on $\mathbb{R}^3$ supported on Σ_e; its trace on Σ_e is the induced surface-area measure of section 3.1. The quantity $\mu_e(\Sigma_e)$ is the surface area of the feasible cone portion. To compute the local surface element, choose any orthonormal frame (p_e, q_e, u_e) with $p_e, q_e \perp u_e$. The cone surface is parametrised away from its excluded apex by

$$\Psi_e(\ell, \xi) = a_e + \ell w_e(\xi), \qquad \ell > 0, \quad \xi \in [0, 2\pi),$$

where

$$w_e(\xi) = \lambda_e u_e + \sqrt{1-\lambda_e^2}\big(\cos\xi\, p_e + \sin\xi\, q_e\big).$$

Since $w_e(\xi)$ has unit length, ℓ is the Euclidean distance from the apex. The derivatives of the parametrisation are

$$\partial_\ell \Psi_e = w_e(\xi), \qquad \partial_\xi \Psi_e = \ell\sqrt{1-\lambda_e^2}\big(-\sin\xi\, p_e + \cos\xi\, q_e\big).$$

These derivative directions are orthogonal, with norms 1 and $\ell\sqrt{1-\lambda_e^2}$, respectively. Hence

$$\|\partial_\ell \Psi_e \times \partial_\xi \Psi_e\| = \sqrt{1-\lambda_e^2}\,\ell.$$

The expressions for the surface element and the bound below depend on the frame only through u_e (via λ_e) and the apex a_e; the auxiliary directions p_e, q_e drop out. The map $\Psi_e : (0,\infty) \times [0, 2\pi) \to C_e$ is a one-to-one parametrisation of C_e. Indeed, $\ell = \|\Psi_e(\ell,\xi) - a_e\|$ is recoverable from the image, and the direction $(\Psi_e(\ell,\xi) - a_e)/\ell$ determines a unique $\xi \in [0, 2\pi)$ because $\sqrt{1-\lambda_e^2} > 0$. Conversely, every $r \in C_e$ can be written in this form with $\ell = \|r - a_e\| > 0$, because its unit direction has inner product λ_e with u_e. By the area formula

[47], applied to this C^1 parametrisation, the pullback of the induced H^2-surface measure has density

$$\sqrt{1-\lambda_e^2}\,\ell$$

with respect to $d\ell\, d\xi$. Equivalently, on the parametrised cone surface,

$$d\mu_e = \sqrt{1-\lambda_e^2}\,\ell\, d\ell\, d\xi,$$

with integration restricted to the preimage of Σ_e. It remains to show that the feasible surface has finite surface area. Since Ω is bounded by construction (*section* 3.1.1), define

$$R_e := \sup_{r\in\Omega} \|r - a_e\| < \infty.$$

If $\Psi_e(\ell, \xi) \in \Sigma_e$, then

$$\ell = \|\Psi_e(\ell, \xi) - a_e\| \le R_e.$$

Therefore Σ_e is contained in the truncated parametrised cone with $0 < \ell \le R_e$, and

$$\mu_e(\Sigma_e) \le \int_0^{2\pi}\int_0^{R_e} \sqrt{1-\lambda_e^2}\,\ell\, d\ell\, d\xi = \pi\sqrt{1-\lambda_e^2}\,R_e^2 < \infty.$$

Thus every feasible surface has finite induced surface-area measure. For the realised event index $m \in E$, the usability criterion in section 3.1 gives $\mu_m(\Sigma_m) > 0$. Hence $0 < \mu_m(\Sigma_m) < \infty$, and the normalised surface-area law Q_m in eq. (3.3) is a well-defined probability measure supported on Σ_m.

A.2 Dirichlet marginalisation and finiteness of the occupancy target

With the notation of section 3.3, introduce auxiliary voxel weights

$$p = (p_v)_{v\in V} \in \Delta_V := \left\{ p_v > 0 \text{ for all } v \in V,\ \sum_{v\in V} p_v = 1 \right\}.$$

For a state $\mathbf{s} \in \mathcal{S}$, the occupancy-only likelihood kernel is

$$L(\mathbf{s} \mid p) := \prod_{v\in V} p_v^{n_v(\mathbf{s})}.$$

This is a likelihood kernel for the event-indexed state, not for an unordered occupancy vector, so no multinomial count factor is included. Equip p with the symmetric Dirichlet prior with parameter $\alpha > 0$ [34],

$$\mathrm{Dir}_\alpha(dp) = \frac{\Gamma(|V|\alpha)}{\Gamma(\alpha)^{|V|}} \prod_{v\in V} p_v^{\alpha-1}\, dp,$$

where dp denotes the standard $(|V|-1)$-dimensional Lebesgue measure on the simplex, expressed in any coordinate chart obtained by omitting one component. Marginalising the auxiliary weights gives

$$\int_{\Delta_V} L(\mathbf{s} \mid p)\, \mathrm{Dir}_\alpha(dp) = \frac{\Gamma(|V|\alpha)}{\Gamma(\alpha)^{|V|}} \int_{\Delta_V} \prod_{v\in V} p_v^{n_v(\mathbf{s})+\alpha-1}\, dp.$$

Using the standard Dirichlet integral [34],

$$\int_{\Delta_V} \prod_{v\in V} p_v^{n_v(\mathbf{s})+\alpha-1}\, dp = \frac{\prod_{v\in V}\Gamma\big(n_v(\mathbf{s})+\alpha\big)}{\Gamma\big(\sum_{v\in V} n_v(\mathbf{s}) + |V|\alpha\big)}.$$

Since $\sum_{v\in V} n_v(\mathbf{s}) = N$, the marginal kernel is

$$\frac{\Gamma(|V|\alpha)}{\Gamma(N+|V|\alpha)} \prod_{v\in V} \frac{\Gamma\big(n_v(\mathbf{s})+\alpha\big)}{\Gamma(\alpha)}.$$

The prefactor and the factors $\Gamma(\alpha)^{-1}$ are independent of $\mathbf{s}$. They are therefore absorbed into the normalising constant in eq. (3.1), yielding the target kernel

$$g_\alpha(\mathbf{s}) := \prod_{v\in V} \Gamma\big(n_v(\mathbf{s})+\alpha\big).$$

The maps $n_v : \mathcal{S} \to \{0,\dots,N\}$ are Borel-measurable, because each B_v is a Borel set in $\mathbb{R}^3$ and each n_v is a finite sum of indicators of the Borel sets $\{x_e \in B_v\} \subseteq \Sigma_e$. It remains to show that this kernel has a finite positive normalising constant. Since E and V are finite and $\sum_{v\in V} n_v(\mathbf{s}) = N$, each count satisfies $0 \le n_v(\mathbf{s}) \le N$. Hence g_α takes values in the finite set

$$\left\{ \prod_{v\in V} \Gamma(k_v+\alpha) : k_v \in \{0,\dots,N\},\ \sum_{v\in V} k_v = N \right\}.$$

Every value in this set is finite and strictly positive because $\alpha > 0$. Therefore there exist constants $0 < c_\alpha \le C_\alpha < \infty$ such that

$$c_\alpha \le g_\alpha(\mathbf{s}) \le C_\alpha \qquad \text{for all } \mathbf{s} \in \mathcal{S}.$$

By appendix A.1, each surviving event has finite positive surface-area measure. Thus the product measure satisfies

$$0 < \mu(\mathcal{S}) = \prod_{e\in E} \mu_e(\Sigma_e) < \infty.$$

The normalising constant

$$Z_\alpha := \int_{\mathcal{S}} g_\alpha(\mathbf{s})\, \mu(d\mathbf{s})$$

therefore obeys

$$0 < c_\alpha\, \mu(\mathcal{S}) \le Z_\alpha \le C_\alpha\, \mu(\mathcal{S}) < \infty.$$

Hence the density $\pi_\alpha = Z_\alpha^{-1} g_\alpha$ in eq. (3.1) is a well-defined probability density with respect to μ. For the baseline case $\alpha = 1$, the identity $\Gamma(k+1) = k!$ gives

$$g_1(\mathbf{s}) = \prod_{v\in V} n_v(\mathbf{s})!.$$

This is the baseline factorial target kernel stated in section 3.3.

A.3 Detailed balance of the single-event Metropolis–Hastings kernel

With the notation of sections 3.3 to 3.5, fix an event $m \in E$. The argument below is the standard reversible-kernel argument for a symmetric Metropolis–Hastings proposal [31, 32, 35]. Because $\alpha > 0$, the kernel $g_\alpha(\mathbf{s}) = \prod_{v \in V} \Gamma\big(n_v(\mathbf{s}) + \alpha\big)$ is strictly positive on $\mathcal{S}$; hence $\pi_\alpha > 0$ on $\mathcal{S}$, and the ratio $\pi_\alpha(z,y)/\pi_\alpha(z,x)$ used below is well defined. For $\mathbf{s} \in \mathcal{S}$ and $y \in \Sigma_m$, write

$$\tau_m(\mathbf{s}, y)$$

for the state obtained by replacing only the m-th component of $\mathbf{s}$ by y. Thus, if $\mathbf{s} = (x_e)_{e \in E}$, then

$$\tau_m(\mathbf{s}, y)_e = \begin{cases} y, & e = m, \\ x_e, & e \neq m. \end{cases}$$

The component kernel for a realised event index m is

$$P_m(\mathbf{s}, A) = \int_{\Sigma_m} a(\mathbf{s}, \tau_m(\mathbf{s}, y))\, \mathbf{1}_A(\tau_m(\mathbf{s}, y))\, Q_m(dy) + r_m(\mathbf{s}) \mathbf{1}_A(\mathbf{s}),$$

where

$$r_m(\mathbf{s}) := 1 - \int_{\Sigma_m} a(\mathbf{s}, \tau_m(\mathbf{s}, y))\, Q_m(dy).$$

Since $a \in [0,1]$ and Q_m is a probability measure, $r_m(\mathbf{s}) \in [0,1]$, and therefore $P_m(\mathbf{s}, \cdot)$ is a probability measure on $\mathcal{S}$. The full one-step kernel is the uniform mixture

$$P(\mathbf{s}, A) := \frac{1}{N} \sum_{m \in E} P_m(\mathbf{s}, A).$$

It is enough to prove detailed balance for each P_m, because a convex mixture of kernels reversible with respect to the same target is reversible with respect to that target. Fix m and decompose the product measure as

$$\mu = \mu_{-m} \otimes \mu_m, \qquad \mu_{-m} := \bigotimes_{e \in E \setminus \{m\}} \mu_e.$$

Under this decomposition, $\mathcal{S}$ is identified with $\big(\prod_{e \neq m} \Sigma_e\big) \times \Sigma_m$ as a product Borel space, and $\mu = \mu_{-m} \otimes \mu_m$ on the product σ-algebra. We write $\mathbf{s} = (z, x)$ accordingly, with $z \in \prod_{e \neq m} \Sigma_e$ containing the unchanged components and $x \in \Sigma_m$ the current m-th component. Since $m \in E$, the surface-measure construction gives $0 < \mu_m(\Sigma_m) < \infty$. The uniform surface-area law therefore has density

$$Q_m(dy) = \frac{1}{\mu_m(\Sigma_m)}\, \mu_m(dy).$$

All integrands below are non-negative and measurable, and the measures involved are finite. By Tonelli's theorem [48], the product integral representation is then justified. The non-rejection part of the detailed-balance measure is therefore, for measurable $A, B \subseteq \mathcal{S}$,

$$\begin{aligned}
&\int_A \pi_\alpha(\mathbf{s}) \int_{\Sigma_m} \mathbf{1}_B(\tau_m(\mathbf{s}, y))\, a(\mathbf{s}, \tau_m(\mathbf{s}, y))\, Q_m(dy)\, \mu(d\mathbf{s}) \\
&= \frac{1}{\mu_m(\Sigma_m)} \int \mathbf{1}_A(z, x)\mathbf{1}_B(z, y)\pi_\alpha(z, x) a\big((z, x), (z, y)\big)\, \mu_{-m}(dz)\, \mu_m(dx)\, \mu_m(dy).
\end{aligned}$$

Here and below, the product integral is over $\prod_{e \neq m} \Sigma_e \times \Sigma_m \times \Sigma_m$. By eq. (3.4), $q_m(x \mid y) = q_m(y \mid x) = 1/\mu_m(\Sigma_m)$, so the proposal ratio in eq. (3.7) is one. Hence

$$a\big((z, x), (z, y)\big) = \min\left\{1, \frac{\pi_\alpha(z, y)}{\pi_\alpha(z, x)}\right\}.$$

Thus

$$\pi_\alpha(z, x) a\big((z, x), (z, y)\big) = \min\{\pi_\alpha(z, x), \pi_\alpha(z, y)\}.$$

Substituting this identity gives

$$\frac{1}{\mu_m(\Sigma_m)} \int \mathbf{1}_A(z, x)\mathbf{1}_B(z, y) \min\{\pi_\alpha(z, x), \pi_\alpha(z, y)\}\, \mu_{-m}(dz)\, \mu_m(dx)\, \mu_m(dy).$$

This expression is unchanged when x and y are interchanged, except that A and B exchange roles. Therefore the accepted-move part satisfies

$$\begin{aligned}
&\int_A \pi_\alpha(\mathbf{s}) \int_{\Sigma_m} \mathbf{1}_B(\tau_m(\mathbf{s}, y))\, a(\mathbf{s}, \tau_m(\mathbf{s}, y))\, Q_m(dy)\, \mu(d\mathbf{s}) \\
&= \int_B \pi_\alpha(\mathbf{s}) \int_{\Sigma_m} \mathbf{1}_A(\tau_m(\mathbf{s}, y))\, a(\mathbf{s}, \tau_m(\mathbf{s}, y))\, Q_m(dy)\, \mu(d\mathbf{s}).
\end{aligned}$$

The rejection part is supported on the diagonal and is therefore symmetric:

$$\int_A \pi_\alpha(\mathbf{s}) r_m(\mathbf{s}) \mathbf{1}_B(\mathbf{s})\, \mu(d\mathbf{s}) = \int_B \pi_\alpha(\mathbf{s}) r_m(\mathbf{s}) \mathbf{1}_A(\mathbf{s})\, \mu(d\mathbf{s}).$$

Combining the accepted-move and rejection parts gives

$$\int_A \pi_\alpha(\mathbf{s}) P_m(\mathbf{s}, B)\, \mu(d\mathbf{s}) = \int_B \pi_\alpha(\mathbf{s}) P_m(\mathbf{s}, A)\, \mu(d\mathbf{s}).$$

Thus P_m satisfies detailed balance with respect to $\pi_\alpha(\mathbf{s})\mu(d\mathbf{s})$. Averaging over the uniformly selected event index preserves this identity, so the full one-step kernel P satisfies

$$\int_A \pi_\alpha(\mathbf{s}) P(\mathbf{s}, B)\, \mu(d\mathbf{s}) = \int_B \pi_\alpha(\mathbf{s}) P(\mathbf{s}, A)\, \mu(d\mathbf{s}).$$

Taking $B = \mathcal{S}$ shows that the probability measure $\pi_\alpha(\mathbf{s})\mu(d\mathbf{s})$ is invariant under P.

This appendix establishes detailed balance and invariance of the idealised single-event Metropolis–Hastings kernel. It does not prove irreducibility, aperiodicity, or convergence from arbitrary initial states. The reconstruction runs are therefore interpreted through the finite execution protocol and the diagnostics reported in sections 3.6 and 4.5.5, not through a separate ergodicity theorem.

A.4 Cone–box clipping and admission certificates

With the notation fixed in sections 4.2.2 and 4.2.3, fix an event e and an azimuth ξ. For $c \in \{x, y, z\}$, let L_c and U_c denote the lower and upper VOI bounds in coordinate c, and let $a_{e,c}$ and $w_{e,c}(\xi)$ denote the corresponding components of a_e and $w_e(\xi)$. The generator ray

$$r_{e,\xi}(\ell) = a_e + \ell w_e(\xi), \qquad \ell > 0,$$

intersects the VOI interior at radius ℓ exactly when

$$L_c < a_{e,c} + \ell w_{e,c}(\xi) < U_c \qquad \text{for all } c \in \{x, y, z\}.$$

For one coordinate, the strict slab condition defines the radial interval

$$J_c^\circ(\xi) := \begin{cases} \left(\dfrac{L_c - a_{e,c}}{w_{e,c}(\xi)}, \dfrac{U_c - a_{e,c}}{w_{e,c}(\xi)} \right), & w_{e,c}(\xi) > 0, \\ \left(\dfrac{U_c - a_{e,c}}{w_{e,c}(\xi)}, \dfrac{L_c - a_{e,c}}{w_{e,c}(\xi)} \right), & w_{e,c}(\xi) < 0, \\ \mathbb{R}, & w_{e,c}(\xi) = 0 \text{ and } a_{e,c} \in (L_c, U_c), \\ \varnothing, & w_{e,c}(\xi) = 0 \text{ and } a_{e,c} \notin (L_c, U_c). \end{cases}$$

The strict radial support at azimuth ξ is

$$I_e^\circ(\xi) = (0, \infty) \cap J_x^\circ(\xi) \cap J_y^\circ(\xi) \cap J_z^\circ(\xi).$$

If any coordinate interval is empty, then $I_e^\circ(\xi) = \varnothing$.

To make the sampled gate a total predicate on every sampled azimuth, assign endpoints also to the degenerate coordinate cases. For nonempty proper intervals, write

$$J_c^\circ(\xi) = \big(\ell_c^{\circ,-}(\xi), \ell_c^{\circ,+}(\xi)\big).$$

Use the extended-real conventions

$$J_c^\circ(\xi) = \mathbb{R} \quad \Longrightarrow \quad \ell_c^{\circ,-}(\xi) = -\infty, \qquad \ell_c^{\circ,+}(\xi) = +\infty,$$

and

$$J_c^\circ(\xi) = \varnothing \quad \Longrightarrow \quad \ell_c^{\circ,-}(\xi) = +\infty, \qquad \ell_c^{\circ,+}(\xi) = -\infty.$$

The positive-ray intersection is then represented by the lower endpoint

$$\ell_e^{\circ,-}(\xi) = \max\big\{0, \ell_x^{\circ,-}(\xi), \ell_y^{\circ,-}(\xi), \ell_z^{\circ,-}(\xi)\big\},$$

and the upper endpoint

$$\ell_e^{\circ,+}(\xi) = \min\big\{\ell_x^{\circ,+}(\xi), \ell_y^{\circ,+}(\xi), \ell_z^{\circ,+}(\xi)\big\}.$$

With this convention, an empty coordinate interval forces $\ell_e^{\circ,-}(\xi) \geq \ell_e^{\circ,+}(\xi)$. Hence the endpoint test is valid without a separate nonemptiness assumption:

$$I_e^\circ(\xi) \neq \varnothing \iff \ell_e^{\circ,-}(\xi) < \ell_e^{\circ,+}(\xi).$$

This strict endpoint inequality is the certificate used by the sampled admission gate.

The sampled gate evaluates this certificate only on the midpoint azimuth grid $\Xi^{(K_{\mathrm{az}})}$ from section 4.2.2. Its event predicate is

$$\widehat{U}_e^{K_{\mathrm{az}}} = \mathbf{1}\left\{\exists\, \xi \in \Xi^{(K_{\mathrm{az}})} : \ell_e^{\circ,-}(\xi) < \ell_e^{\circ,+}(\xi)\right\}.$$

If $\widehat{U}_e^{K_{\mathrm{az}}} = 1$, then some sampled azimuth ξ has a radius $\ell \in I_e^\circ(\xi)$. Hence

$$a_e + \ell w_e(\xi) \in \Omega^\circ.$$

The witness has positive radius and is therefore not the cone apex. Under the nondegenerate cone convention fixed in chapter 3, the cone surface is regular at the witness. Since Ω° is open, a relative surface neighbourhood of the witness remains inside Ω°. By the surface-measure convention in appendix A.1, this neighbourhood has positive surface-area measure. Therefore the event has strictly positive feasible surface area, and

$$\widehat{U}_e^{K_{\mathrm{az}}} = 1 \implies e \in E.$$

Consequently,

$$\widehat{E}_{K_{\mathrm{az}}}^{\mathrm{gate}} \subseteq E.$$

For surface-area accumulation and proposal construction, use the corresponding closed clipping interval. Whenever $I_e^\circ(\xi) \neq \varnothing$, set

$$[\ell_e^-(\xi), \ell_e^+(\xi)] := [\ell_e^{\circ,-}(\xi), \ell_e^{\circ,+}(\xi)].$$

The strict interval supplies the admission and initialisation witness. The closed interval supplies the same surface-area mass, because the added radial endpoints and cone–VOI boundary pieces are μ_e-null under the surface-measure convention.

This null-boundary statement follows from the cone parametrisation. For a coordinate face $x_c = h$, the preimage under $\Psi_e(\ell, \xi) = a_e + \ell w_e(\xi)$ is given by

$$a_{e,c} + \ell g_c(\xi) = h, \qquad g_c(\xi) := w_{e,c}(\xi) = \omega_{e,c}^0 + \omega_{e,c}^{\cos} \cos\xi + \omega_{e,c}^{\sin} \sin\xi,$$

where

$$\omega_{e,c}^0 = \lambda_e u_{e,c}, \qquad \omega_{e,c}^{\cos} = s_e p_{e,c}, \qquad \omega_{e,c}^{\sin} = s_e q_{e,c}.$$

Where $g_c(\xi) \neq 0$, this set is the graph $\ell = (h - a_{e,c})/g_c(\xi)$ and is one-dimensional in the (ℓ, ξ)-domain. Where $g_c(\xi) = 0$, the face can contribute only azimuths satisfying $a_{e,c} = h$. Since g_c cannot vanish identically, the zero-denominator face case consists of finitely many generator rays. These sets are $d\mu_e$-null because

$$d\mu_e = \ell s_e \, d\ell \, d\xi.$$

The finite union over the six VOI faces is therefore μ_e-null.

A.5 Surface-area-uniform proposal on a clipped cone

Andreyev et al. explicitly identify the computational issue that motivates this appendix: rejection-based generation of random half-cone locations becomes inefficient when only a small part of the cone intersects the VOI, and more efficient sampling of the VOI-intersecting cone portion is left for future work [20]. The cited SOE literature does not, to our knowledge, give an explicit clipped-cone inverse-CDF construction for the corresponding surface-area-uniform proposal. This appendix records the construction used in the implementation and verifies the proposal law needed by the Metropolis–Hastings kernel.

Fix a usable event $e \in E$. By definition,

$$\mu_e(\Sigma_e) > 0.$$

The clipping construction in appendix A.4 gives, for each azimuth ξ, either an empty clipped radial set or a closed radial interval

$$[\ell_e^-(\xi), \ell_e^+(\xi)].$$

By the endpoint convention established there, these closed intervals define the same induced surface-area law as the corresponding strict-interior intervals.

With the same parametrisation and $s_e = \sin\theta_e > 0$, the surface-measure calculation in appendix A.1 gives

$$d\mu_e = \ell s_e \, d\ell \, d\xi.$$

A surface-area-uniform proposal on the clipped cone must therefore sample (ℓ, ξ) on the clipped parameter domain with density proportional to ℓs_e.

The azimuth decomposition isolates visible azimuth segments on which the active radial bounds are fixed. For each coordinate $c \in \{x, y, z\}$, use the denominator g_c defined above. The lower and upper coordinate faces have offsets

$$\tau_{c,L} = L_c - a_{e,c}, \qquad \tau_{c,U} = U_c - a_{e,c},$$

and therefore define the candidate radii

$$\ell_{c,L}(\xi) = \frac{\tau_{c,L}}{g_c(\xi)}, \qquad \ell_{c,U}(\xi) = \frac{\tau_{c,U}}{g_c(\xi)}.$$

On intervals where $g_c(\xi) > 0$, $\ell_{c,L}$ is an entry candidate and $\ell_{c,U}$ is an exit candidate. On intervals where $g_c(\xi) < 0$, their roles are reversed. If $g_c(\xi) = 0$, the coordinate is constant along the ray. Under the closed clipping convention, that coordinate slab contributes no radial constraint exactly when

$$L_c \le a_{e,c} \le U_c, \qquad \text{equivalently} \qquad \tau_{c,L} \le 0 \le \tau_{c,U}.$$

Otherwise the azimuth is empty. The ray-domain constraint contributes the lower-bound candidate 0. A face entry candidate with value exactly 0 does not replace this ray-domain lower endpoint; only a strictly positive entry can become an active lower face. Thus a zero-offset face coincident with $\ell = 0$ is represented as $\ell_e^-(\xi) = 0$, not as a weighted lower face term.

The visible-azimuth partition is obtained by cutting $[0, 2\pi)$ at the non-identically-zero denominator roots and at crossovers between non-identical face candidates. For faces $f_i = (c_i, h_i)$, with $h_i \in \{L, U\}$ and $\tau_{f_i} := \tau_{c_i,h_i}$, a crossover satisfies

$$\tau_{f_1} g_{c_2}(\xi) - \tau_{f_2} g_{c_1}(\xi) = 0.$$

Each denominator-root or crossover equation has the form

$$D + E\cos\xi + F\sin\xi = 0.$$

Identically true equations do not create breakpoints. All remaining equations have finitely many roots on $[0, 2\pi)$. Since there are only six coordinate faces and three coordinate denominators, the breakpoint set $\mathcal{B}_e \subset [0, 2\pi)$, with angles represented modulo 2π, is finite.

If $\mathcal{B}_e = \varnothing$, the partition contains one full-circle visible azimuth segment with stored endpoints

$$\xi_\sigma^- = 0, \qquad \xi_\sigma^+ = 2\pi,$$

classified at representative azimuth π. If $\mathcal{B}_e = \{\beta_1 < \cdots < \beta_J\}$ with $J \geq 1$, the partition contains the visible azimuth segments on (β_j, β_{j+1}) for $j < J$ and the wrapped visible azimuth segment with stored endpoints

$$\xi_\sigma^- = \beta_J, \qquad \xi_\sigma^+ = \beta_1.$$

Each visible azimuth segment is classified at an interior representative azimuth, reduced modulo 2π when the representative lies on a wrapped branch. This classification fixes the denominator sign pattern, empty/nonempty status, active lower candidate, and active upper face on the segment.

Let $\mathcal{P}_e$ denote the retained finite collection of positive-area visible azimuth segments. For each $\sigma \in \mathcal{P}_e$, define the lifted branch

$$\sigma^\uparrow = \left(\xi_\sigma^{\uparrow,-}, \xi_\sigma^{\uparrow,+}\right) := \begin{cases} (\xi_\sigma^-, \xi_\sigma^+), & \xi_\sigma^+ > \xi_\sigma^-, \\ (\xi_\sigma^-, \xi_\sigma^+ + 2\pi), & \xi_\sigma^+ \leq \xi_\sigma^-. \end{cases}$$

The no-breakpoint full-circle segment is stored as $(0, 2\pi)$ and is therefore not a wrapped segment. All segment integrals and inverse-CDF evaluations use the lifted coordinate on $\sigma^\uparrow$; sampled azimuths are reduced modulo 2π only after the lifted coordinate has been drawn. Zero-area arcs and boundary rays are discarded because they have zero induced surface-area measure.

For $\xi \in \sigma^\uparrow$, with the trigonometric functions read periodically in ξ, the upper endpoint has the form

$$\ell_e^+(\xi) = \frac{\tau_{\sigma,+}}{g_{\sigma,+}(\xi)}, \qquad g_{\sigma,+}(\xi) = A_{\sigma,+} + B_{\sigma,+}\cos\xi + C_{\sigma,+}\sin\xi.$$

The lower endpoint is either the ray-domain endpoint,

$$\ell_e^-(\xi) = 0,$$

or another active face constraint,

$$\ell_e^-(\xi) = \frac{\tau_{\sigma,-}}{g_{\sigma,-}(\xi)}, \qquad g_{\sigma,-}(\xi) = A_{\sigma,-} + B_{\sigma,-}\cos\xi + C_{\sigma,-}\sin\xi.$$

The segment construction includes all zeros of the coordinate denominators $w_{e,c}(\xi)$ and all cross-face equalities among candidate radial constraints in the breakpoint set. Hence, on each retained open segment, the denominator sign pattern, empty/nonempty status, active lower constraint, and active upper constraint are fixed. For every active face term $\tau/g(\xi)$ used on such a segment, g has no zero in the segment interior.

The endpoint convention for the analytic integrals is improper and one-sided. Endpoint zeros of inactive denominators are irrelevant. If an active denominator with nonzero face offset τ had a zero at a segment endpoint and remained active up to that endpoint, the corresponding radial bound $\tau/g(\xi)$ would diverge and the weighted integral $\tau^2 I_g$ would not define a finite segment mass. Such an arc is therefore not a retained finite-mass proposal segment. The lower ray-domain case is represented separately by $\ell_e^-(\xi) = 0$, so no denominator integral is introduced for that endpoint.

The segment masses require the primitive

$$I_g(\xi_a, \xi_b) := \int_{\xi_a}^{\xi_b} \frac{1}{g(\xi)^2}\, d\xi, \qquad g(\xi) = A + B\cos\xi + C\sin\xi,$$

on intervals where g has no zero. If $R = 0$, then $g(\xi) = A$ and

$$I_g(\xi_a, \xi_b) = \frac{\xi_b - \xi_a}{A^2}.$$

Assume otherwise that

$$R = \sqrt{B^2 + C^2}, \qquad \phi = \operatorname{atan2}(C, B), \qquad t = \xi - \phi, \qquad u = \tan\frac{t}{2},$$

and define

$$A_0 = A + R, \qquad B_0 = A - R, \qquad \Delta = A_0 B_0 = A^2 - B^2 - C^2.$$

On a lifted interval that does not cross the half-angle pole $t = \pi \pmod{2\pi}$, the primitive increment is

$$I_g(\xi_a, \xi_b) = J(u_b) - J(u_a), \qquad u_a = \tan\frac{\xi_a - \phi}{2}, \qquad u_b = \tan\frac{\xi_b - \phi}{2}.$$

The antiderivative branch is selected by Δ. For $\Delta > 0$,

$$J(u) = -\frac{2R}{\Delta}\frac{u}{A_0 + B_0 u^2} + \frac{2A}{\Delta\sqrt{\Delta}}\arctan\left(\frac{u\sqrt{\Delta}}{A_0}\right).$$

For $\Delta < 0$, set $\delta = \sqrt{-\Delta}$ and use

$$J(u) = -\frac{2R}{\Delta}\frac{u}{A_0 + B_0 u^2} - \frac{2A}{\delta^3} H\left(\frac{\delta u}{A_0}\right),$$

where

$$H(x) = \begin{cases} \operatorname{artanh}(x), & |x| < 1, \\ \operatorname{acoth}(x), & |x| > 1, \end{cases} \qquad \operatorname{acoth}(x) = \frac{1}{2} \log \left| \frac{x+1}{x-1} \right|.$$

For $\Delta = 0$, the two nonzero cases are

$$J(u) = \frac{1}{2A^2} \left(u + \frac{u^3}{3} \right), \qquad B_0 = 0 \quad (A = R),$$

and

$$J(u) = -\frac{1}{2A^2} \left(\frac{1}{u} + \frac{1}{3u^3} \right), \qquad A_0 = 0 \quad (A = -R).$$

At a half-angle pole $t = \pi \pmod{2\pi}$, the primitive is evaluated by one-sided limits. The left limit corresponds to $u \to +\infty$, and the right limit corresponds to $u \to -\infty$. Thus, if a lifted integration endpoint lies at a half-angle pole, $J(u)$ is interpreted as the one-sided limit from the side of the integration interval.

If a lifted integration interval crosses a half-angle pole, the interval is split at the pole and the two one-sided limits are used. With the principal arctangent in the $\Delta > 0$ branch, this is equivalently implemented by adding

$$\frac{2|A|\pi}{\Delta\sqrt{\Delta}}$$

for each crossed half-angle pole. In the $\Delta < 0$ branch, the one-sided limits at $u = \pm\infty$ are both zero, so no branch-jump correction is added. In the $\Delta = 0,\ A_0 = 0$ branch, the one-sided limits at $u = \pm\infty$ are also zero. The remaining $\Delta = 0,\ B_0 = 0$ branch has $g(\xi) = 0$ at the half-angle pole, so it cannot occur on a lifted integration interval on which g has no zero. These conventions make I_g reproducible from the face coefficients (A, B, C), including endpoint-pole and wrapped-segment cases.

Let $\chi_\sigma = 0$ when the lower endpoint on segment σ is the ray-domain endpoint, including the zero-offset face tie described above. Let $\chi_\sigma = 1$ only when the lower endpoint is an active face constraint with positive radial entry on the segment interior. After radial marginalisation, the surface-area density in azimuth is

$$m_\sigma(\xi) = \frac{s_e}{2} \left[\left(\ell_e^+(\xi) \right)^2 - \left(\ell_e^-(\xi) \right)^2 \right].$$

In the following formula, $I_{g_{\sigma,\pm}}$ denotes the finite improper one-sided integral over the lifted segment endpoints whenever the corresponding weighted face term is present. Segments for which this finite mass condition fails are excluded from $\mathcal{P}_e$. Using the fixed active constraints on σ, the exact segment mass is

$$M_\sigma = \frac{s_e}{2} \left[\tau_{\sigma,+}^2 I_{g_{\sigma,+}} \left(\xi_\sigma^{\uparrow,-}, \xi_\sigma^{\uparrow,+} \right) - \chi_\sigma \tau_{\sigma,-}^2 I_{g_{\sigma,-}} \left(\xi_\sigma^{\uparrow,-}, \xi_\sigma^{\uparrow,+} \right) \right].$$

Only segments with $M_\sigma > 0$ enter $\mathcal{P}_e$. The total clipped surface mass is

$$M_e := \sum_{\sigma \in \mathcal{P}_e} M_\sigma.$$

The retained visible azimuth segments form a finite partition, up to μ_e-null boundaries, of the parameter preimage of Σ_e under Ψ_e. Therefore

$$M_e = \int_{\Sigma_e} d\mu_e = \mu_e(\Sigma_e),$$

and the fixed event assumption gives

$$M_e > 0.$$

Thus the segment probabilities are well defined.

The proposal first selects a segment with probability

$$\mathbb{P}(\sigma) = \frac{M_\sigma}{M_e}, \qquad \sigma \in \mathcal{P}_e.$$

A cumulative-mass implementation may use any fixed ordering of $\mathcal{P}_e$ with half-open cumulative intervals; cumulative boundaries have probability zero under the continuous mass draw and do not change the law.

Conditionally on segment σ, the lifted azimuth is sampled by inverting the partial segment mass

$$C_\sigma(\eta) = \frac{s_e}{2}\left[\tau_{\sigma,+}^2 I_{g_{\sigma,+}}\left(\xi_\sigma^{\uparrow,-}, \eta\right) - \chi_\sigma \tau_{\sigma,-}^2 I_{g_{\sigma,-}}\left(\xi_\sigma^{\uparrow,-}, \eta\right)\right], \qquad \eta \in \sigma^\uparrow.$$

This function satisfies

$$C_\sigma(\xi_\sigma^{\uparrow,-}) = 0, \qquad C_\sigma(\xi_\sigma^{\uparrow,+}) = M_\sigma.$$

On a retained positive-area segment, $\ell_e^+(\eta) > \ell_e^-(\eta)$ except possibly at segment endpoints, and $s_e > 0$. Hence $C'_\sigma(\eta) = m_\sigma(\eta) > 0$ in the segment interior, so C_σ is continuous and strictly increasing from 0 to M_σ. The inverse is therefore well defined on $(0, M_\sigma)$.

Drawing $V_\xi \sim \mathrm{Unif}(0,1)$ and solving

$$C_\sigma(\eta) = V_\xi M_\sigma$$

gives a lifted azimuth $\eta \in \sigma^\uparrow$. The sampled azimuth on the canonical domain is

$$\xi = \eta \bmod 2\pi.$$

For every Borel set $D \subset [0, 2\pi)$, the conditional azimuth law is

$$\mathbb{P}(\xi \in D \mid \sigma) = \frac{1}{M_\sigma} \int_{D_\sigma^\uparrow} m_\sigma(\eta)\, d\eta,$$

where

$$D_\sigma^\uparrow := \{\eta \in \sigma^\uparrow : \eta \bmod 2\pi \in D\}.$$

Thus the integral is taken on the selected lifted branch before reduction modulo 2π.

Given the sampled lifted azimuth η, set $\xi = \eta \bmod 2\pi$ and write

$$\ell_- = \ell_e^-(\xi), \qquad \ell_+ = \ell_e^+(\xi).$$

The conditional radial density is proportional to ℓ on $[\ell_-, \ell_+]$. Drawing $V_\ell \sim \mathrm{Unif}(0,1)$ and setting

$$\ell = \sqrt{\ell_-^2 + V_\ell(\ell_+^2 - \ell_-^2)}$$

gives this law. For $\rho \in [\ell_-, \ell_+]$,

$$\mathbb{P}(\ell \leq \rho \mid \xi) = \frac{\rho^2 - \ell_-^2}{\ell_+^2 - \ell_-^2},$$

and hence

$$f_{\ell|\xi}(\rho) = \frac{2\rho}{\ell_+^2 - \ell_-^2}.$$

This equals the radial density induced by the surface-area element, since

$$\frac{\rho s_e}{\frac{s_e}{2}(\ell_+^2 - \ell_-^2)} = \frac{2\rho}{\ell_+^2 - \ell_-^2}.$$

Combining segment selection, conditional azimuth inversion, and radial sampling gives the joint density

$$f_e(\xi, \ell) = \frac{\ell s_e}{M_e}$$

on the clipped parameter domain and zero outside it. Since $M_e = \mu_e(\Sigma_e)$ and $d\mu_e = \ell s_e \, d\ell \, d\xi$ under Ψ_e, the pushforward of this joint law is the normalised induced surface-area measure on Σ_e. The local proposal density with respect to μ_m is therefore the chapter 3 within-event density

$$q_m(x'_m \mid x_m) = \frac{1}{\mu_m(\Sigma_m)}, \qquad x'_m \in \Sigma_m.$$

For the derivation above, $m = e$.

Appendix B

Software Documentation and Repository Structure

The software developed for this work is provided in the public repository `soe-fn`. The repository contains a Python implementation of the Stochastic Origin Ensemble (SOE) reconstruction core for fast-neutron double-scatter cone-event data. Its purpose is to provide a reproducible implementation of the reconstruction pipeline used in this thesis, from read-only event ingestion to voxelised reconstruction output.

The code is organized around a single reconstruction workflow. Input events are loaded from ng-imager HDF5 files, converted into canonical cone-event objects, filtered according to the configured event-validity rules, and combined with an externally specified volume of interest (VOI). The resulting `ReconstructionInput` object is then passed to the SOE reconstruction chain. The chain places one representative point per admitted event on its cone surface, voxelises the current ensemble state, and evolves the state using a Metropolis–Hastings sampler. The retained states are finally accumulated into a mean occupancy image, while the terminal state and metadata sidecars are written as reconstruction artifacts.

B.1 Repository Layout

The repository follows a source-layout Python package structure. The main implementation is located under `src/soe/`, while executable reconstruction entry points and example configuration files are placed under `orchestration/`. Public example input data are stored under `data/`, and regression tests are stored under `tests/`.

The most important top-level files and directories are:

```
soe-fn/
  README.md
  ARCH.md
  pyproject.toml
  requirements-dev.txt
  orchestration/
  data/
```

```
src/soe/
tests/
```

The file `README.md` gives the basic installation and execution instructions. The file `ARCH.md` documents the internal architecture and the main dependency boundaries of the reconstruction code. The package metadata and runtime dependencies are defined in `pyproject.toml`; the public runtime depends only on `numpy` and `h5py`, with development tools listed separately in `requirements-dev.txt`.

B.2 Core Package Structure

The `src/soe/` package is divided into focused subpackages. The `contracts` module defines the core data contracts, including the VOI configuration, canonical event object, registration transform, filtering configuration, and reconstruction input object. The `core` module assembles canonical events and VOI configuration into the reconstruction input used by the rest of the pipeline.

The `adapters` package contains boundary code for external data formats. In particular, `adapters/hdf5_ngimager` implements read-only HDF5 event ingestion, while `adapters/voi_voxel` implements VOI and voxel-grid utilities. Geometry-related routines are collected under `geometry`; these include cone geometry, ray–box intersection logic, sampled surrogate checks, and exact bounded-box surface geometry used by the proposal mechanism.

The live SOE reconstruction chain is implemented under `soe`. This subpackage contains the reconstruction state, occupancy calculations, proposal logic, target-density ratio calculations, Metropolis–Hastings kernel, run protocol, estimators, and chain-health summaries. The `analysis` subpackage contains the higher-level reconstruction runner and artifact helpers used to persist reconstruction results.

B.3 Configuration and Execution

Reconstructions are driven by TOML configuration files. The main command-line entry point is:

```
python3 orchestration/run_reconstruction.py --config path/to/reconstruction.toml
```

Example configurations are provided under `orchestration/configs/`. These configurations specify the input HDF5 file, VOI bounds, voxel-grid shape, run length, optional filtering and registration parameters, deterministic seeds, and output artifact directory. Relative paths inside a configuration file are resolved from the location of that configuration file.

A typical full reconstruction writes the following artifacts:

```
retained_mean.npy
terminal_state.npy
reconstruction_metadata.json
reconstruction_diagnostics.json
chain_health.json
```

The retained mean stores the reconstructed mean occupancy image. The terminal state stores the final voxelised ensemble state. The metadata and diagnostics JSON files record the reconstruction geometry, artifact provenance, run metrics, and chain-health information needed to interpret the numerical output.

B.4 Public Example Data

The repository includes small public HDF5 inputs under `data/`. These are used by the example reconstruction configurations and provide reproducible test cases for the public code. The public data layout includes toy gamma and neutron cases, as well as water-phantom neutron datasets at several energies. Generated outputs are intentionally written to ignored local artifact directories so that runtime products are not committed to the repository.

B.5 Testing and Reproducibility

The repository includes a regression test suite under `tests/`. The tests cover the public reconstruction contracts, HDF5 read-only boundary, VOI and voxel semantics, cone geometry, bounded-box proposal geometry, SOE state initialisation, Metropolis–Hastings kernel behaviour, reconstruction artifact writing, and orchestration entry points. The test suite can be executed with:

```
python3 -m pytest -q
```

The public release branch was verified with 348 collected tests and 348 passing tests. This provides a reproducibility check that the released repository contains the intended SOE reconstruction core and that the public examples, configuration surface, and reconstruction pipeline remain internally consistent.